INCH-POUND

MIL-M-28787/356
29 March 1991

MILITARY SPECIFICATION

MODULES, DIGITAL, STANDARD ELECTRONIC

SUPPORT, MPU COMMUNICATIONS MODULE, KEY CODE AHH

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic support, MPU communications module. This module is class I as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification, standards and handbook. The following specification, standards and handbook form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

HANDBOOK

MILITARY

MIL-HDBK-217 - Reliability Prediction of Electronic Equipment.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a failure rate predicted in accordance with MIL-STD-1389. The predicted failure rate shall not exceed 0.9432 failures per million operating hours when predicted in accordance with MIL-HDBK-217.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be AHH.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 100 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

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TABLE I. Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
A1	Isolation, initial	II	S	10	INF	10	INF	10	INF	MOhm
	end-of-life	II	Q	1	INF	1	INF	1	INF	MOhm
A2-A14	Isolation, initial	II	X	10	INF	10	INF	10	INF	MOhm
	end-of life	II	Q	1	INF	1	INF	1	INF	MOhm
B1-B4	Continuity	III	X	0.0	1.0	0 0	1.0	0.0	1.0	Ohm
B5-B8	Continuity	III	Q	0.0	0.1	0.0	0.1	0.0	0.1	Ohm
C	Voo stress	IV	S	--------Module shall pass all remaining tests----------						
D1-D7	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	nA
D8-D40	Ibd	V	S	0.0	100.0	0.0	100.0	0.0	100.0	uA
D41,D42	Ibd	V	S	0.0	1.10	0.0	1.10	0.0	1.10	mA
E1-E42	Vio	VI	S	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	V
F1-F7	Ios	VII	S	-100.0	-15.0	-100.0	-15.0	-100.0	-15.0	nA
F8-F11	Ios	VII	S	-130.0	-15.0	-130.0	-15.0	-130.0	-15.0	nA
F12-F15	Ios	VII	S	-110.0	-15.0	-110.0	-15.0	-110.0	-15.0	nA
F16	Ioe	VII	S	-130.5	-30.1	-130.5	-30.1	-130.5	-30.1	nA
F17-F20	Ios	VII	S	-130.0	-30.0	-130.0	-30.0	-130.0	-30.0	nA
F21-F36	Ios	VII	S	-250.0	-30.0	-250.0	-30.0	-250.0	-30.0	nA
G1-G15	Vol	VIII	X	0.0	0.4	0.0	0.4	0.0	0.4	V
G16-G37	Vol	VIII	X	0.0	0.5	0.0	0.5	0.0	0.5	V
H1-H15	Voh	IX	X	2.5	4.5	2.5	4.5	2.5	4.5	V
H16-H36	Voh	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	V
H37,H38	Voh	IX	X	2.4	2.6	2.4	2.6	2.4	2.6	V

-See footnotes at end of table

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
J1-J4	Iil	X	X	-786.8	-485.7	-786.8	-485.7	-786.8	-485.7	uA
J5-J7	Iil	X	X	-250.0	0.0	-250.0	0.0	-250.0	0.0	uA
J8-J20	Iil	X	X	-936.8	-585.7	-936.8	-585.7	-936.8	-585.7	uA
J21-J36	Iil	X	X	-936.8	-515.7	-936.8	-515.7	-936.8	-515.7	uA
J37-J39	Iil	X	X	-400.0	-100.0	-400.0	-100.0	-400.0	-100.0	uA
J40	Iil	X	X	-440.0	-100.0	-440.0	-100.0	-440.0	-100.0	uA
J41,J42	Iil	X	X	-650.0	-30.0	-650.0	-30.0	-650.0	-30.0	uA
K1-K4	Iih	XI	X	-351.3	-295.2	-351.3	-295.2	-351.3	-295.2	uA
K5-K7	Iih	XI	X	0.0	25.0	0.0	25.0	0.0	25.0	uA
K8-K36	Iih	XI	X	-314.7	-266.7	-314.7	-266.7	-314.7	-266.7	uA
K37-K40	Iih	XI	X	0.0	20.0	0.0	20.0	0.0	20.0	uA
K41,K42	Iih	XI	X	0.0	45.0	0.0	45.0	0.0	45.0	uA
L	Icc	XII	X	57.3	572.7	57.3	572.7	57.3	572.7	mA
M1-M7	Propagation delay time (tPLH)	XIII	S	5	44	5	60	5	60	nS
M8-M15	Propagation delay time (tPLH)	XIII	S	2	27	2	37	2	37	nS
M16-M23	Propagation delay time (tPLH)	XIII	S	2	32	2	47	2	47	nS
M24-M46	Propagation delay time (tPLH)	XIII	S	5	52	5	52	5	52	nS
M47-M62	Propagation delay time (tPLH)	XIII	S	0	27	0	32	0	32	nS

-See footnotes at end of table

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
N1-N7	Propagation delay time (tPHL)	XIV	S	5	52	5	73	5	73	nS
N8-N15	Propagation delay time (tPHL)	XIV	S	2	31	2	41	2	41	nS
N16-N23	Propagation delay time (tPHL)	XIV	S	2	31	2	46	2	46	nS
N24-N46	Propagation delay time (tPHL)	XIV	S	5	56	5	56	5	56	nS
N47-N62	Propagation delay time (tPHL)	XIV	S	0	28	0	33	0	33	nS
P1-P16	Setup time (tsu)	XV	S	20	--	20	--	20	--	nS
Q1-Q16	Hold time (tH)	XV	S	5	--	5	--	5	--	nS
R1-R3	Clock frequency (Fmax)	XV	S	--	10	--	10	--	10	MHz
R4-R6	Minimum pulse width (PW)	XV	S	20	--	20	--	20	--	nS
R7	Minimum pulse width (PW)	XV	S	25	--	25	--	25	--	nS

-See footnotes at end of table

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
-S	Functional	XVI	X	See table XVI						
T1-T27	Output crosstalk (low level)	XVII	Q	0	400	0	400	0	400	mV
T28-T35	Output crosstalk (low level)	XVII	Q	0	500	0	500	0	500	mV
V1-V37	Output crosstalk (high level)	XVIII	Q	0	400	0	400	0	400	mV
W	PSTA	XIX	Q	0	400	0	400	0	400	mV
X	PSFC	XX	S	4.125	6.875	4.125	6.875	4.125	6.875	uF
Y1-Y4	Cin	XXI	Q	2.0	55.0	2.0	55.0	2.0	55.0	pF
Y5-Y7	Cin	XXI	Q	2.0	40.0	2.0	40.0	2.0	40.0	pF
Y8-Y20	Cin	XXI	Q	2.0	55.0	2.0	55.0	2.0	55.0	pF
Y21-Y39	Cin	XXI	Q	2.0	40.0	2.0	40.0	2.0	40.0	pF
Y40	Cin	XXI	Q	2.0	30.0	2.0	30.0	2.0	30.0	pF
Y41,Y42	Cin	XXI	Q	2.0	70.0	2.0	70.0	2.0	70.0	pF
Z	Threshold	XXII	X	See table XXII						

-See footnotes on next page.

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/356-1. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. The module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.0 ± 0.5 V dc, except the output short circuit current limits apply only for a power supply voltage of 5.5 V dc ± 1 percent.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

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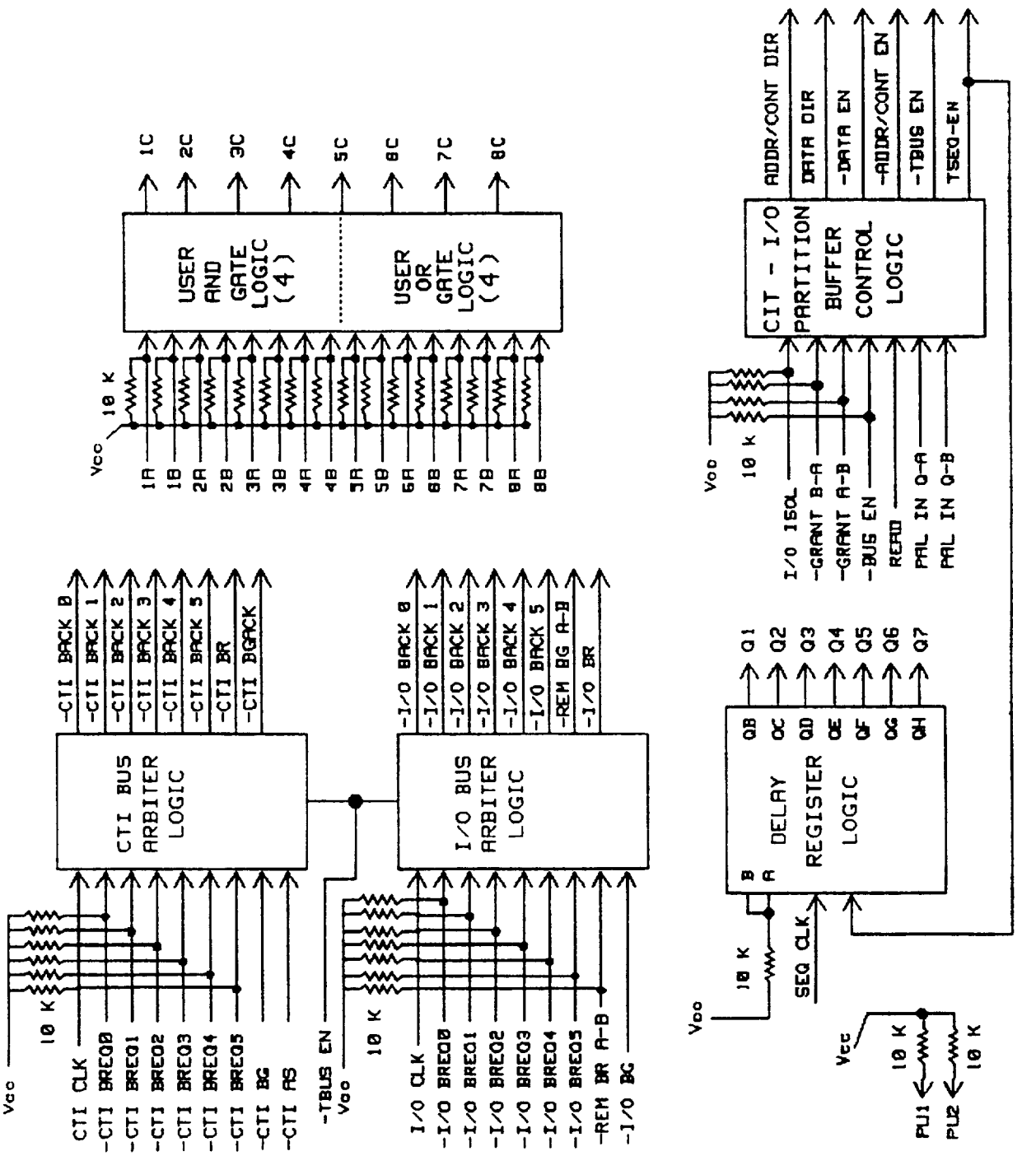


FIGURE 1. AHH functional block diagram.

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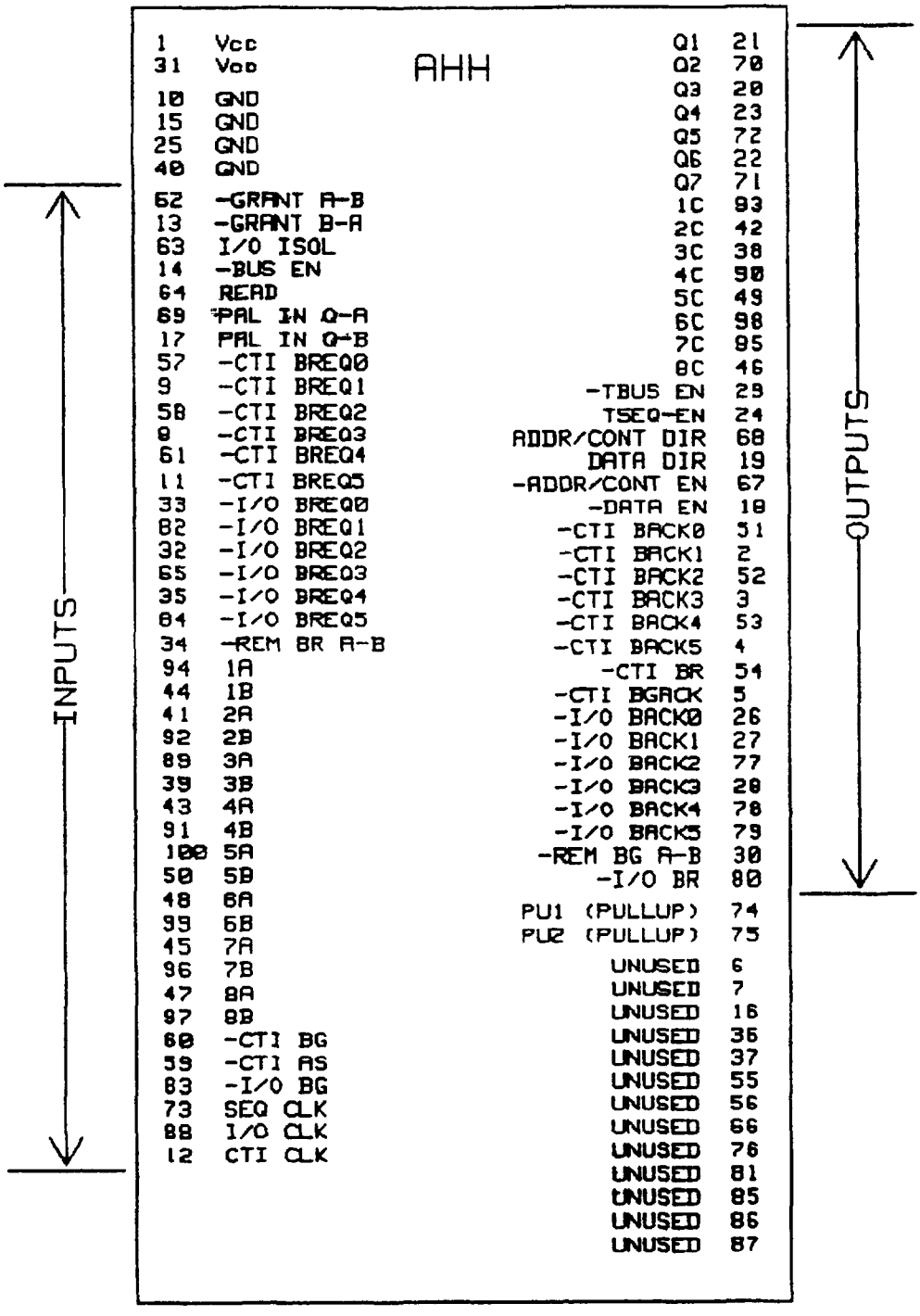


FIGURE 2. AHH pin assignment.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through Z shall be performed in any order with the following exceptions. Test conditions A, C, D, E and F are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C, D, E or F are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E, and F where applicable, followed by the remaining tests in the sequence.

4.4.3 Multiple testing of pins. Singular pins listed multiple times in a test table indicate multiple devices connected to the pins. The module shall be tested so that each device connected to a pin is exercised and tested.

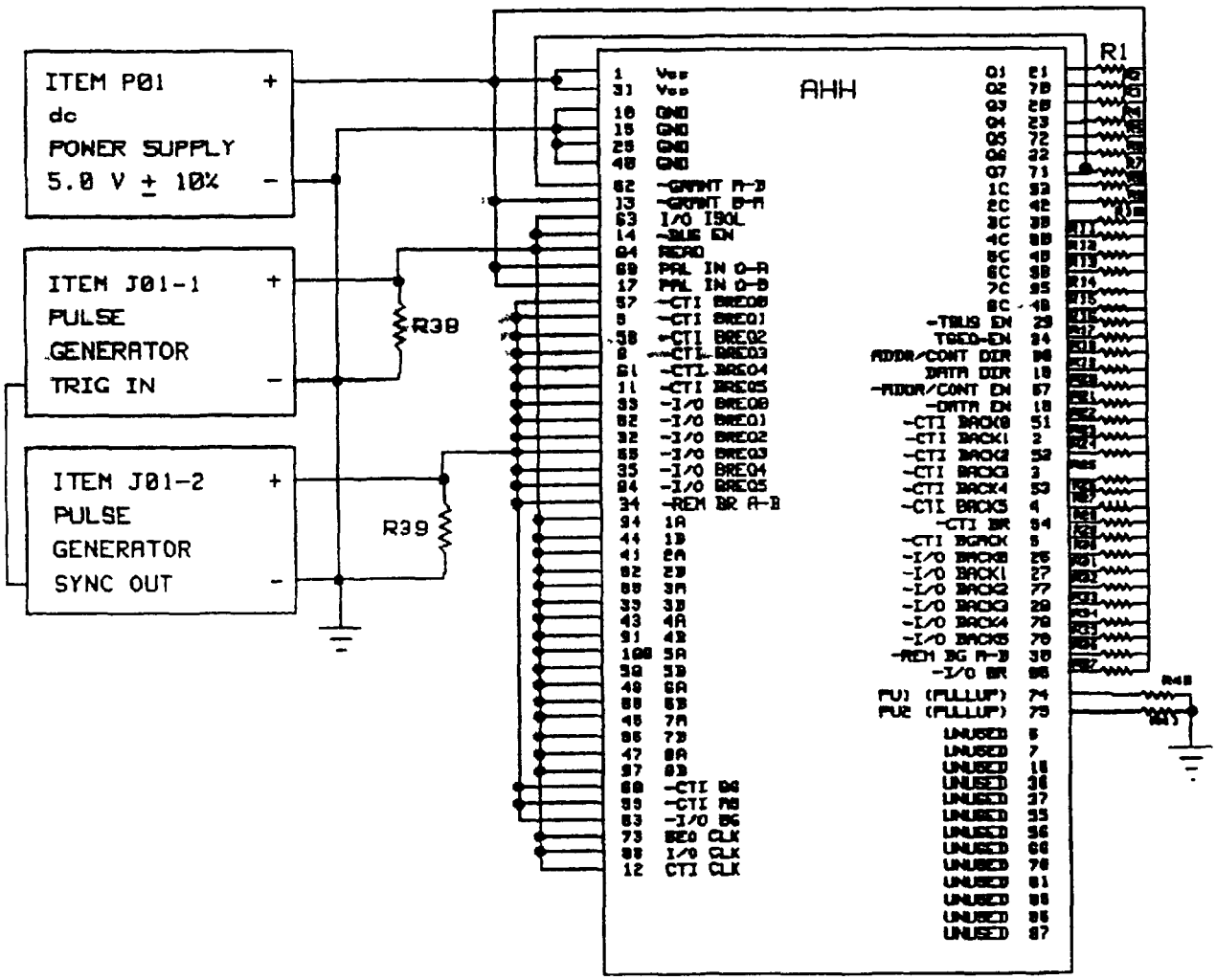
4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through Z in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, G through X and Z shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated as shown on figures 3 and 4.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 6. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition X, shall be performed using the initial limits at 25 ± 5°C ambient.

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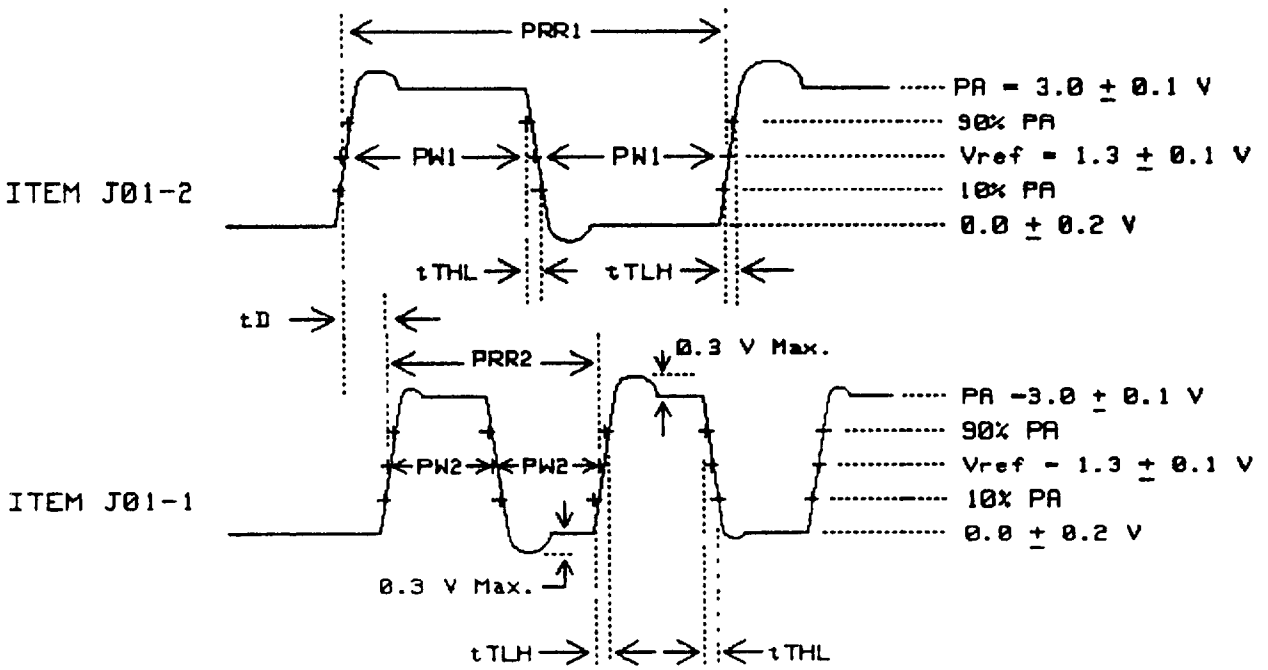
NOTES:

1. Resistor values:

   R1-R15 = 1.27 KOhm ± 1 percent
   R18-R21 = 626 Ohm ± 1 percent
   R17 = 657 Ohm ± 1 percent
   R16, R22-R37 = 417 Ohm ± 1 percent
   R38,R39 = 51 Ohm ± 5 percent
   R40,R41 = 17.6 KOhm ± 1 percent

FIGURE 3. AHH operating temperature and life test circuit.

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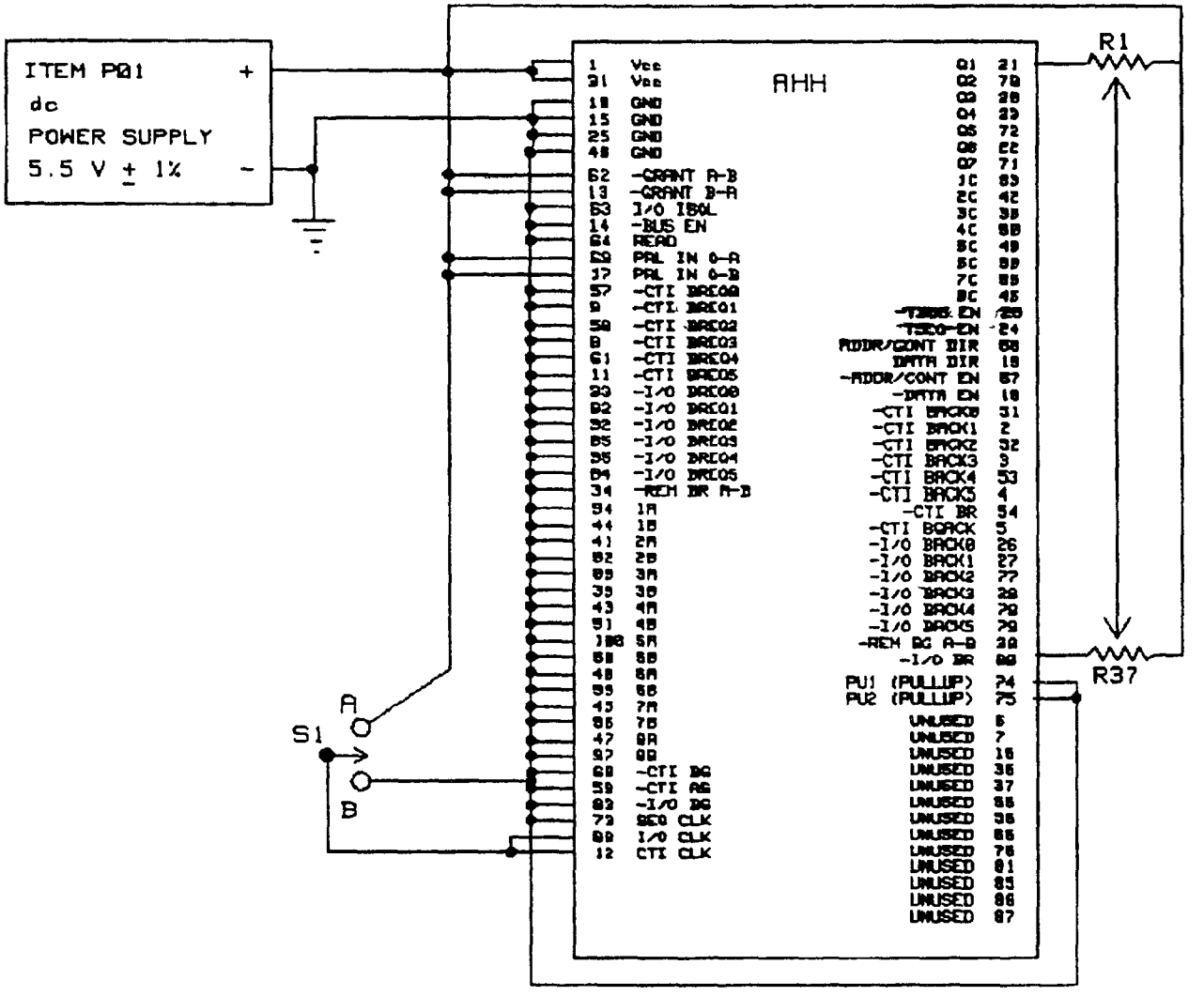


NOTES:

1. PRR1 = 500 kHz ± 20 percent
   PW1 = 1000 nS ± 100 nS
   PRR2 = 1.0 MHz ± 20 percent
   PW2 = 500 ± 100 nS
   tTLH = tTHL = 10 ± 2 nS
   tD = 30 ± 5 nS

FIGURE 4. Operating temperature and life test waveforms.

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NOTES:

1. Resistor values:
   R1-R15 = 1.27 KOhm ± 1 percent
   R18-R21 = 626 Ohm ± 1 percent
   R17 = 657 Ohm ± 1 percent
   R16, R22-R37 = 417 Ohm ± 1 percent

2. S1= DPST switch

3. Alternate S1 from position B to position A two times, pausing a minimum of 100 ns after each move.

FIGURE 5. AHH component temperature test circuit.

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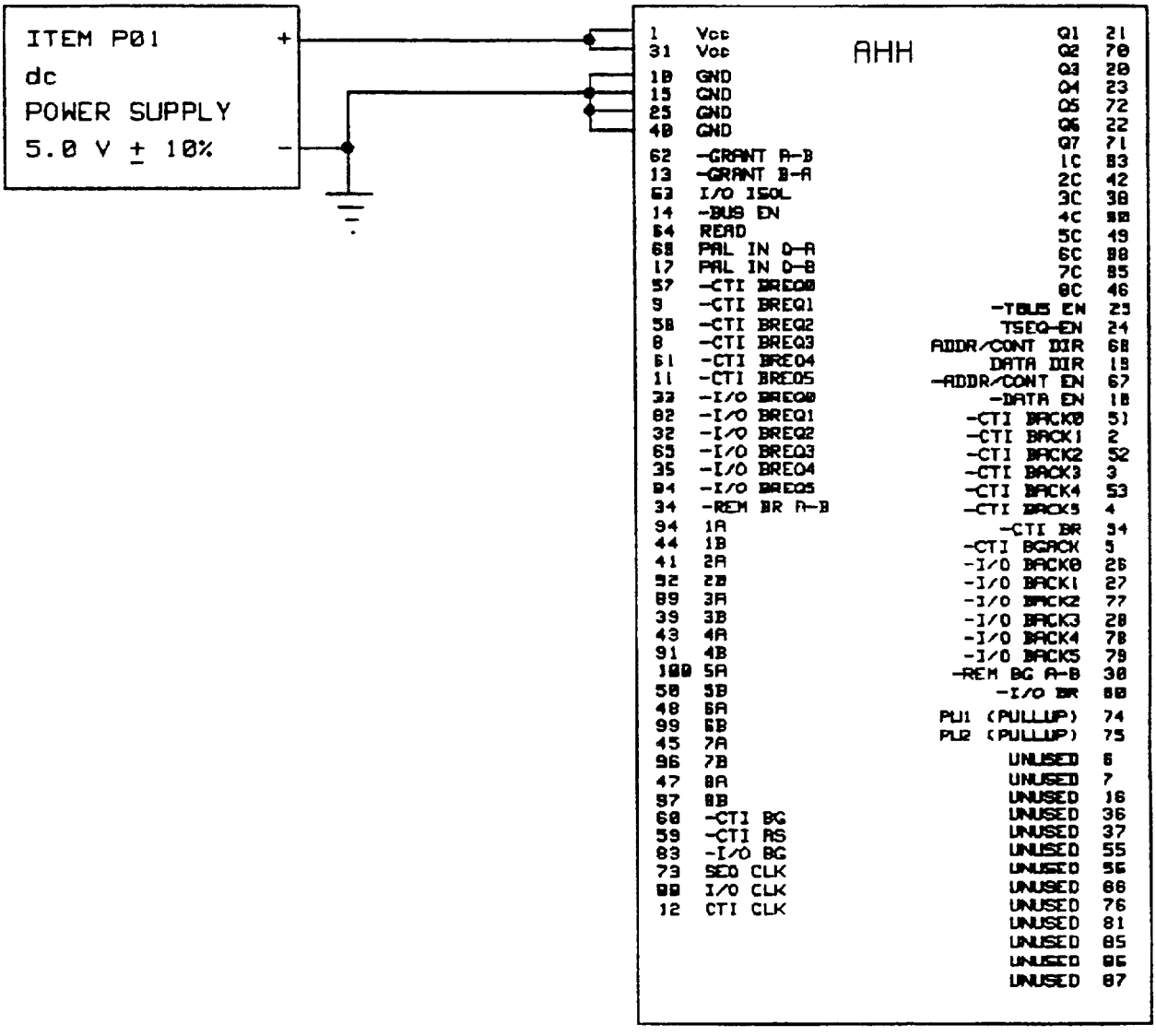


FIGURE 6. AHH vibration test circuit.

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4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figures 3 and 4. Following the life test and while still at the maximum class temperature, electrical tests B, G through X and Z shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, G through X and Z shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Cout = output capacitance.
(c) Ibd = input breakdown current.
(d) Icc = power supply current.
(e) Iih = input high level current.
(f) Iil = input low level current.
(g) Iin = input forcing current.
(h) Ioh = output high level current.
(i) Iol = output low level current.
(j) Ios = output short circuit current.
(k) Iozh = output high level leakage current.
(l) Iozl = output low level leakage current.
(m) NA = not applicable.
(n) PSFC = power supply filter capacitance.
(o) PSTA = power supply transient amplitude.
(p) Vic = input clamp diode voltage.
(q) Vih = input high level voltage.
(r) Vil = input low level voltage.
(s) Vin = input forcing voltage.
(t) Voh = output high level voltage.
(u) Vol = output low level voltage.
(v) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 3.19 W (max); 2.0 W (typical).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (uA)	Iih max (uA)	Ibd max (mA)	Cin max (pF)
-GRANT A-B	-1.5	0.8	2.0	-786.8	-295.2	1.0	55.0
-GRANT B-A	-1.5	0.8	2.0	-786.8	-295.2	1.0	55.0
I/O ISOL	-1.5	0.8	2.0	-786.8	-295.2	1.0	55.0
-BUS EN	-1.5	0.8	2.0	-786.8	-295.2	1.0	55.0
READ	-1.5	0.8	2.0	-250.0	25.0	1.0	40.0
PAL IN Q-A	-1.5	0.8	2.0	-250.0	25.0	1.0	40.0
PAL IN Q-B	-1.5	0.8	2.0	-250.0	25.0	1.0	40.0
-CTI BREQ0	-1.5	0.7	2.0	-936.8	-266.7	.100	55.0
-CTI BREQ1	-1.5	0.7	2.0	-936.8	-266.7	.100	55.0
-CTI BREQ2	-1.5	0.7	2.0	-936.8	-266.7	.100	55.0
-CTI BREQ3	-1.5	0.7	2.0	-936.8	-266.7	.100	55.0
-CTI BREQ4	-1.5	0.7	2.0	-936.8	-266.7	.100	55.0
-CTI BREQ5	-1.5	0.7	2.0	-936.8	-266.7	.100	55.0
-I/O BREQ0	-1.5	0.7	2.0	-936.8	-266.7	.100	55.0

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (uA)	Iih max (uA)	Ibd max (mA)	Cin max (pF)
-I/O BREQ1	-1.5	0.7	2.0	-936.8	-266.7	.100	55.0
-I/O BREQ2	-1.5	0.7	2.0	-936.8	-266.7	.100	55.0
-I/O BREQ3	-1.5	0.7	2.0	-936.8	-266.7	.100	55.0
-I/O BREQ4	-1.5	0.7	2.0	-936.8	-266.7	.100	55.0
-I/O BREQ5	-1.5	0.7	2.0	-936.8	-266.7	.100	55.0
-REM BR A-B	-1.5	0.7	2.0	-936.8	-266.7	.100	55.0
1A	-1.5	0.7	2.0	-936.8	-266.7	.100	40.0
1B	-1.5	0.7	2.0	-936.8	-266.7	.100	40.0
2A	-1.5	0.7	2.0	-936.8	-266.7	.100	40.0
2B	-1.5	0.7	2.0	-936.8	-266.7	.100	40.0
3A	-1.5	0.7	2.0	-936.8	-266.7	.100	40.0
3B	-1.5	0.7	2.0	-936.8	-266.7	.100	40.0
4A	-1.5	0.7	2.0	-936.8	-266.7	.100	40.0
4B	-1.5	0.7	2.0	-936.8	-266.7	.100	40.0

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (uA)	Iih max (uA)	Ibd max (mA)	Cin max (pF)
5A	-1.5	0.7	2.0	-936.8	-266.7	.100	40.0
5B	-1.5	0.7	2.0	-936.8	-266.7	.100	40.0
6A	-1.5	0.7	2.0	-936.8	-266.7	.100	40.0
6B	-1.5	0.7	2.0	-936.8	-266.7	.100	40.0
7A	-1.5	0.7	2.0	-936.8	-266.7	.100	40.0
7B	-1.5	0.7	2.0	-936.8	-266.7	.100	40.0
8A	-1.5	0.7	2.0	-936.8	-266.7	.100	40.0
8B	-1.5	0.7	2.0	-936.8	-266.7	.100	40.0
-CTI BG	-1.5	0.7	2.0	-400.0	20.0	.100	40.0
-CTI AS	-1.5	0.7	2.0	-400.0	20.0	.100	40.0
-I/O BG	-1.5	0.7	2.0	-400.0	20.0	.100	40.0
SEQ CLK	-1.5	0.7	2.0	-440.0	20.0	.100	30.0
I/O CLK	-1.5	0.7	2.0	-650.0	45.0	1.1	70.0
CTI CLK	-1.5	0.7	2.0	-650.0	45.0	1.1	70.0

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6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
-CTI BACK5	2.4	0.5	-2.0	12.0	NA	NA	-30.0	NA
-CTI BR	2.4	0.5	-2.0	12.0	NA	NA	-30.0	NA
-CTI BGACK	2.4	0.5	-2.0	12.0	NA	NA	-30.0	NA
-I/O BACK0	2.4	0.5	-2.0	12.0	NA	NA	-30.0	NA
-I/O BACK1	2.4	0.5	-2.0	12.0	NA	NA	-30.0	NA
-I/O BACK2	2.4	0.5	-2.0	12.0	NA	NA	-30.0	NA
-I/O BACK3	2.4	0.5	-2.0	12.0	NA	NA	-30.0	NA
-I/O BACK4	2.4	0.5	-2.0	12.0	NA	NA	-30.0	NA
-I/O BACK5	2.4	0.5	-2.0	12.0	NA	NA	-30.0	NA
-REM BG A-B	2.4	0.5	-2.0	12.0	NA	NA	-30.0	NA
-I/O BR	2.4	0.5	-2.0	12.0	NA	NA	-30.0	NA
7C	2.5	0.4	-0.4	4.0	NA	NA	-15.0	NA
8C	2.5	0.4	-0.4	4.0	NA	NA	-15.0	NA

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
-TBUS EN	NA	0.5	NA	7.5	NA	NA	NA	NA
TSEQ-EN	2.4	0.5	-1.98	7.56	NA	NA	-30.1	NA
ADDR/CONT DIR	2.4	0.5	-2.0	8.0	NA	NA	-30.0	NA
DATA DIR	2.4	0.5	-2.0	8.0	NA	NA	-30.0	NA
-ABDR/CONT EN	2.4	0.5	-2.0	8.0	NA	NA	-30.0	NA
-DATA EN	2.4	0.5	-2.0	8.0	NA	NA	-30.0	NA
-CTI BACK0	2.4	0.5	-2.0	12.0	NA	NA	-30.0	NA
-CTI BACK1	2.4	0.5	-2.0	12.0	NA	NA	-30.0	NA
-CTI BACK2	2.4	0.5	-2.0	12.0	NA	NA	-30.0	NA
-CTI BACK3	2.4	0.5	-2.0	12.0	NA	NA	-30.0	NA
-CTI BACK4	2.4	0.5	-2.0	12.0	NA	NA	-30.0	NA
Q1	2.5	0.4	-0.4	4.0	NA	NA	-15.0	NA
Q2	2.5	0.4	-0.4	4.0	NA	NA	-15.0	NA
Q3	2.5	0.4	-0.4	4.0	NA	NA	-15.0	NA

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
Q4	2.5	0.4	-0.4	4.0	NA	NA	-15.0	NA
Q5	2.5	0.4	-0.4	4.0	NA	NA	-15.0	NA
Q6	2.5	0.4	-0.4	4.0	NA	NA	-15.0	NA
Q7	2.5	0.4	-0.4	4.0	NA	NA	-15.0	NA
1C	2.5	0.4	-0.4	4.0	NA	NA	-15.0	NA
2C	2.5	0.4	-0.4	4.0	NA	NA	-15.0	NA
3C	2.5	0.4	-0.4	4.0	NA	NA	-15.0	NA
4C	2.5	0.4	-0.4	4.0	NA	NA	-15.0	NA
5C	2.5	0.4	-0.4	4.0	NA	NA	-15.0	NA
6C	2.5	0.4	-0.4	4.0	NA	NA	-15.0	NA

6.3.3 Dynamic parameters. The dynamic parameters appear with the dynamic test.

6.4 Module function. The module function is printed herein in the appendix.

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6.4.1 Description of I/O pins.

Signal name	Pin number	I/O	Functional description
-GRANT A-B	62	I	Grant bus A to B. This active low input signal indicates that a DMA device in the CTI section has been granted the I/O bus. A 10 KOhm pull-up resistor is located on this signal to assure that if this input is unused, it will be set to the inactive state.
-GRANT B-A	13	I	Grant bus B to A. This active low signal indicates that a DMA device in the I/O section has been granted the CTI bus. A 10 KOhm pull-up resistor is located on this signal to assure that if this input is unused, it will be set to the inactive state.
I/O ISOL	63	I	Input/Output isolate. This active low input signal is used to condition the enabling of Address, Read/Write, Data and Strobe Bidirectional Control Buffers. A 10 KOhm pull-up resistor is located on this signal to assure that if this input is unused, it will be set to the inactive state.
-BUS EN	14	I	Bus enable. This active low input signal is used to enable the address/control and data bidirectional control buffers. A 10 KOhm pull-up resistor is located on this signal to assure that if this input is unused, it will be set to the inactive state.
READ	64	I	This input signal defines the data bus cycle as a read (high) or write (low) cycle.

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6.4.1 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
PAL IN Q-A	69	I	PAL input, - CTI section. This active high input is used to sequence the enabling of the CTI partition bidirectional control buffers.
PAL IN Q-B	17	I	PAL input, - I/O section. This active high input is used to sequence the enabling of the I/O partition bidirectional control buffers.
-CTI BREQ0 -CTI BREQ1 -CTI BREQ2 -CTI BREQ3 -CTI BREQ4 -CTI BREQ5	57 9 58 8 61 11	I I I I I I	CTI bus requests inputs 0-5. Each active low input signal comes from a DMA device telling the AHH that the device wishes control of the bus. -CTI BREQ5 has the highest priority bus request and -CTI BREQ0 the lowest. A 10 KOhm pull-up resistor is located on each of these signals to assure that if these inputs are unused, they will be set to the inactive state.
-I/O BREQ0 -I/O BREQ1 -I/O BREQ2 -I/O BREQ3 -I/O BREQ4 -I/O BREQ5	33 82 32 65 35 84	I I I I I I	I/O DMA bus request 0-5. Each of these active low input signals indicate that I/O DMA bus request is active. -I/O BREQ5 has the highest priority.
-REM BR A-B	34	I	Remote bus request CTI to I/O. This active low input signal is activated by the microprocessor support logic when the microprocessor or a CTI-resident DMA controller requires access to the I/O partition bus. This input signal is the highest priority input to the I/O partition. A 10

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6.4.1 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
			KOhm pull-up resistor is located on this input to assure that if this input is unused, it will be set to the inactive state.
1A, 2A 3A, 4A 1B, 2B 3B, 4B	94,41 89,43 44,92 39,91	I I I I	Logic AND inputs. These inputs are the signals to the four two-input-positive AND gates which are available for general purpose use. A 10 KOhm pull-up resistor is located on each of these inputs to assure that unused inputs are set to the inactive state. Unused inputs should be left open. Tying LS gate inputs together significantly increases the gates input noise susceptibility.
5A, 6A 7A, 8A 5B, 6B 7B, 8B	100,48 45,47 50,99 96,97	I I I I	Logic OR inputs. These inputs are the four two-input-positive OR gates which are available for general purpose use. A 10 KOhm pull-up resistor is located on each of these inputs. Logically unused inputs must be tied to ground.
-CTI BG	60	I	CTI bus grant. This active low input signal comes from the 68000 microprocessor and tells the AHH that the highest priority pending DMA device can now take control of the bus.
-CTI AS	59	I	CTI address strobe. This active low signal indicates that there is a valid address on the system address bus and is used to condition the generation of -CTI BGACK.

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6.4.1 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
-I/O BG	83	I	Input/Output bus grant. This active low input signal is asserted when the microprocessor relinquishes the bus.
SEQ CLK	73	I	Sequencer clock. This input signal continuously clocks the delay register on its rising edge.
I/O CLK	88	I	Input/Output clock. The I/O clock controls the latching of I/O DMA bus requests, external bus request and control signal as well as controlling the outputting of bus acknowledge signals from the arbiter PAL on its rising edge.
CTI CLK	12	I	CTI Clock input. The CTI CLK signal controls the latching of bus request and control signals into the input register (LS377) on its rising edge. The outputting of bus acknowledge and control signals out of the PAL16R8 in the CTI section is also regulated by the CTI clock.
Q1, Q2 Q3, Q4 Q5, Q6 Q7	21,70 20,23 72,22 71	O O O O	8 - bit parallel output of the delay register. These active high outputs are used to delay the enabling of the master bidirectional control buffers after the data bidirectional control buffers are enabled. Q0 is not used.
1C, 2C 3C, 4C	93,42 38,90	O O	Logic AND outputs. These output signals are from the four two-input-positive AND gates.

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6.4.1 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
5C, 6C 7C, 8C	49,98 95,46	O O	Logic OR outputs. These output signals are from the four two-input-positive OR gates.
-TBUS EN	29	O	Test point for bus enable. This active low test point indicates when the CTI and I/O arbiter outputs are enabled. There must be no external connections made to this signal.
TSEQ-EN	24	O	Testpoint timing sequence enable. This active high testpoint is asserted when either -GRANT B-A or -GRANT A-B is active low to enable the register logic to begin sequencing for bidirectional control buffer enables. This output is used for test purposes only. There must be no external connections made to this signal.
ADDR/CONT DIR	68	O	Address/Control direction. This active high output controls the direction of the bidirectional address and control buffers between the CTI and I/O partitions.
DATA DIR	19	O	Data bidirectional control buffer direction. This active high output is used to control the data direction of the bidirectional control buffers between the CTI and I/O partitions.
-ADDR/CONT EN	67	O	Address/Control enable. This active low output enables the bidirectional buffers for the address and control bus between the CTI and I/O partitions.

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6.4.1 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
-DATA EN	18	O	Bidirectional control data enable. This active low output enables the bidirectional data buffers between the CTI and I/O partitions.
-CTI BACK0 -CTI BACK1 -CTI BACK2 -CTI BACK3 -CTI BACK4 -CTI BACK5	51 2 52 3 53 4	O O O O O O	Control Timing and Interface (CTI) bus acknowledge 0-5. These active low output signals are used to notify the appropriate DMA controller in the CTI partition that it is bus master.
-CTI BR	54	O	Control Timing and Interface (CTI) bus request. This active low output signal is asserted to a 68000 controller module whenever the arbiter logic accepts a bus request from the DMA controllers in the CTI partition provided that -CTI BGACK is high.
-CTI BGACK	5	O	Control Timing and Interface (CTI) bus grant acknowledge. This active low output signal is asserted to a 68000 based controller module which then negates the bus request (-CTI BR) in response.
-I/O BACK0 -I/O BACK1 -I/O BACK2 -I/O BACK3 -I/O BACK4 -I/O BACK5	26 27 77 28 78 79	O O O O O O	Control Timing and Interface (CTI) bus acknowledge 0-5. These active low output signals are used to notify the appropriate DMA controller in the CTI partition that it is bus master.
-REM BG A-B	30	O	Remote bus grant CTI to I/O. This active low output signal indicates that a DMA device in the CTI partition has been granted the use of the I/O partition bus.

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6.4.1 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
-I/O BR	80	O	Input/Output bus request. This active low output signal is asserted to the microprocessor whenever the arbiter logic accepts a bus request from the DMA controller in the CTI partition.
PU1	74	O	Pull-up 1. These pullup (10 KOhm) resistors are pinned out for general purpose use.
PU2	75	O	Pull-up 2. These pullup (10 KOhm) resistors are pinned out for general purpose use.

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6.5 Description of tests. The following tables and associated figures describe the tests to be performed.

(a) Isolation test - table II, figure 7.

(b) Continuity test - table III, figure 8.

(c) Vcc stress test - table IV, figure 9.

(d) Ibd test - table V, figure 10.

(e) Vic test - table VI, figure 11.

(f) Ios test - table VII, figure 12.

(g) Vol test - table VIII, figures 13 and 14.

(h) Voh test - table IX, figures 13 and 14.

(i) Iil test - table X, figure 15.

(j) Iih test - table XI, figure 15.

(k) Icc test - table XII, figure 16.

(l) Propagation delay time (tPLH) test - table XIII, figures 17, 18 and 19.

(m) Propagation delay time (tPHL) test - table XIV, figures 17, 18 and 19.

(n) Dynamic test - table XV, figures 20 through 23.

(o) Functional test - table XVI, figure 20.

(p) Output crosstalk tests - tables XVII and XVIII, figures 24 and 25.

(q) PSTA test - table XIX, figures 26 and 27.

(r) PSFC test - table XX, figure 28.

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(s) Cin test - table XXI.

(t) Threshold test - table XXII.

6.6 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Custodians:
Army - ER
Navy - SH
Air Force - 85

Review activities:
Army - AT
Navy - AS, MC
Air Force - 13, 19
DLA - ES

Preparing activity:
Navy - SH

(Project 5963-0055)

Agent: NW

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TABLE II. Isolation test.

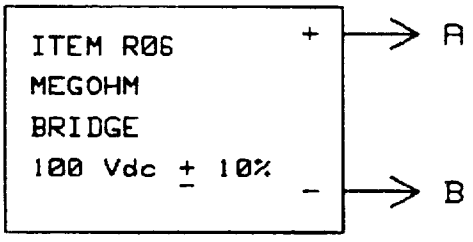
Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
A1	Isolation, frame	All Pins	Frame	10	INF	MOhm	7
A2	Isolation, used to unused	All used pins	6,7,16, 36,37,55, 56,66,76, 81,85,86, 87	10	INF	MOhm	7
A3	Isolation, unused to unused	6	7,16,36, 37,55,56, 66,76,81, 85,86,87	10	INF	MOhm	7
A4	Isolation, unused to unused	7	16,36,37, 55,56,66, 76,81,85, 86,87				
A5	Isolation, unused to unused	16	36,37,55, 56,66,76, 81,85,86, 87				
A6	Isolation, unused to unused	36	37,55,56, 66,76,81, 85,86,87				
A7	Isolation, unused to unused	37	55,56,66, 76,81,85, 86,87				
A8	Isolation, unused to unused	55	56,66,76, 81,85,86, 87				
A9	Isolation, unused to unused	56	66,76,81, 85,86,87	10	INF	MOhm	7

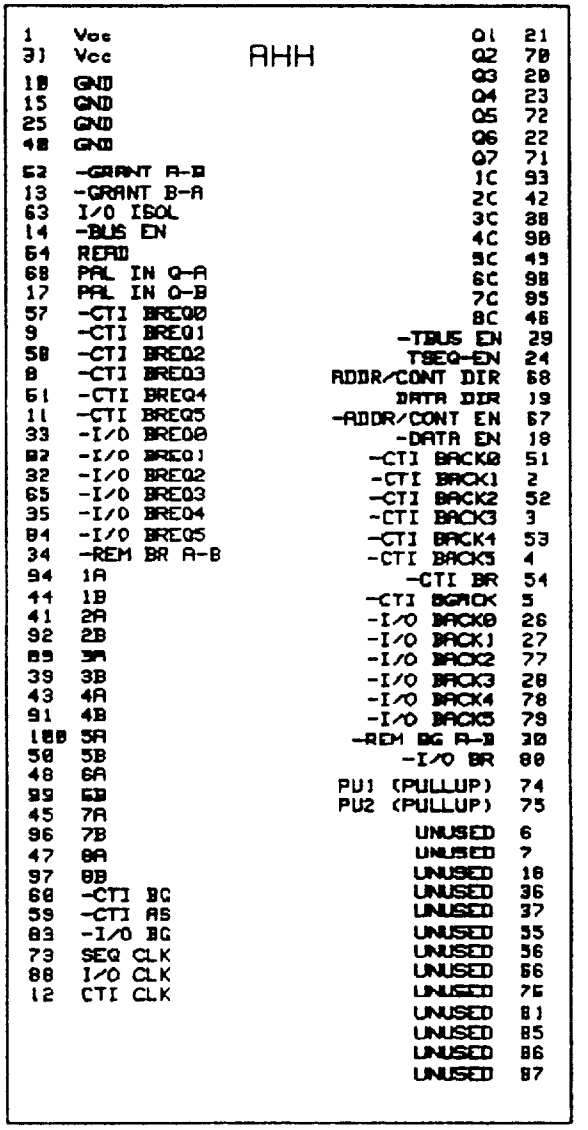
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TABLE II. Isolation test - Continued.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
A10	Isolation, unused to unused	66	76,81,85, 86,87	10	INF	MOhm	7
A11	Isolation, unused to unused	76	81,85,86, 87				
A12	Isolation, unused to unused	81	85,86,87				
A13	Isolation, unused to unused	85	86,87				
A14	Isolation, unused to unused	86	87	10	INF	MOhm	7

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NOTES:

1. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

2. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

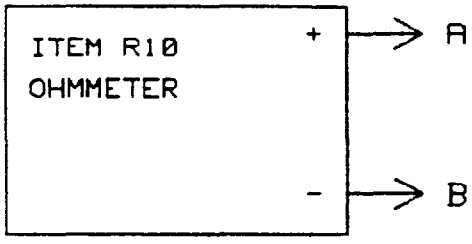
FIGURE 7. Isolation test circuit for table II.

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TABLE III. Continuity test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
B1	Continuity, Vcc	1	31	0.0	1.0	Ohm	8
B2	Continuity, GND	10	15				
B3	Continuity, GND	10	25				
B4	Continuity, GND	10	40	0.0	1.0	Ohm	8
B5	Continuity, Vcc	1	31	0.0	0.1	Ohm	8
B6	Continuity, GND	10	15				
B7	Continuity, GND	10	25				
B8	Continuity, GND	10	40	0.0	0.1	Ohm	8

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AHH

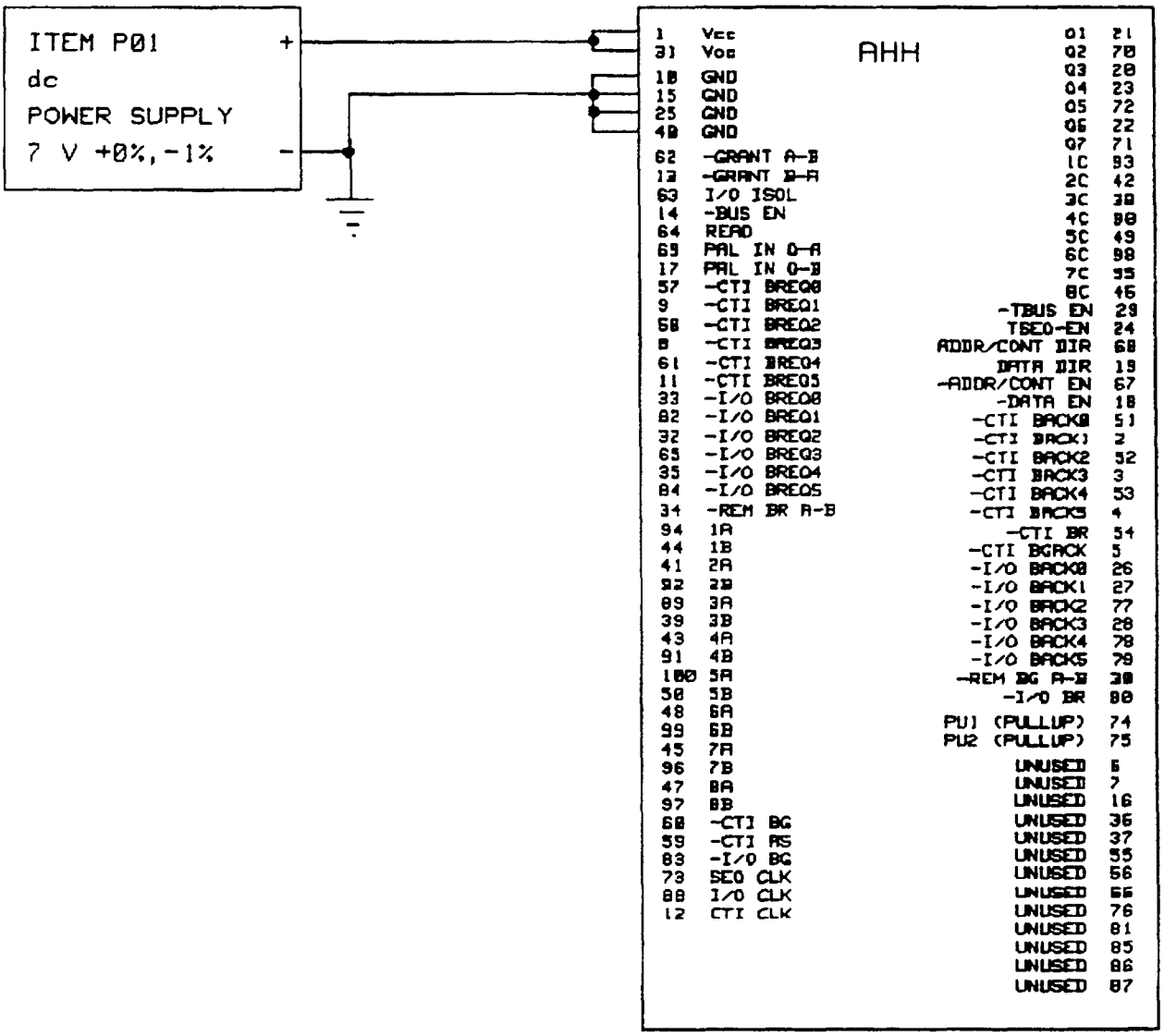
1	Vcc	Q1	21
31	Vcc	Q2	70
10	GND	Q3	20
15	GND	Q4	23
25	GND	Q5	72
40	GND	Q6	22
62	-GRANT A-B	Q7	71
13	-GRANT B-A	1C	93
63	I/O IBOL	2C	42
14	-BUS EN	3C	90
64	READ	4C	50
69	PAL IN Q-A	5C	49
17	PAL IN Q-B	6C	90
67	-CTI BREQ0	7C	95
9	-CTI BREQ1	8C	46
58	-CTI BREQ2	-TBUS EN	29
8	-CTI BREQ3	TSEQ-EN	24
61	-CTI BREQ4	ADDR/CONT DIR	68
11	-CTI BREQ5	DATA DIR	19
33	-I/O BREQ0	-ADDR/CONT EN	67
82	-I/O BREQ1	-DATA EN	18
32	-I/O BREQ2	-CTI BACK0	51
65	-I/O BREQ3	-CTI BACK1	2
35	-I/O BREQ4	-CTI BACK2	52
84	-I/O BREQ5	-CTI BACK3	3
94	-REM BR A-B	-CTI BACK4	53
94	1A	-CTI BACK5	4
44	1B	-CTI BR	54
41	2A	-CTI BGACK	5
52	2B	-I/O BACK0	26
89	3A	-I/O BACK1	27
39	3B	-I/O BACK2	77
43	4A	-I/O BACK3	28
81	4B	-I/O BACK4	78
100	5A	-I/O BACK5	79
50	5B	-REM BG A-B	30
48	6A	-I/O BR	80
99	6B	PU1 (PULLUP)	74
45	7A	PU2 (PULLUP)	75
96	7B	UNUSED	6
47	8A	UNUSED	7
87	8B	UNUSED	16
60	-CTI BG	UNUSED	36
59	-CTI AS	UNUSED	37
83	-I/O BG	UNUSED	55
73	SEQ CLK	UNUSED	56
88	I/O CLK	UNUSED	86
12	CTI CLK	UNUSED	76
		UNUSED	81
		UNUSED	85
		UNUSED	86
		UNUSED	87

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TABLE IV. Power supply stress test.

Test cond	Power supply	25 ± 5°C Initial limits	Fig
C	Vcc	Module shall pass all remaining tests.	9

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NOTES:

1. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

FIGURE 9. Vcc stress test circuit for table IV.

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TABLE V. Input breakdown current test.

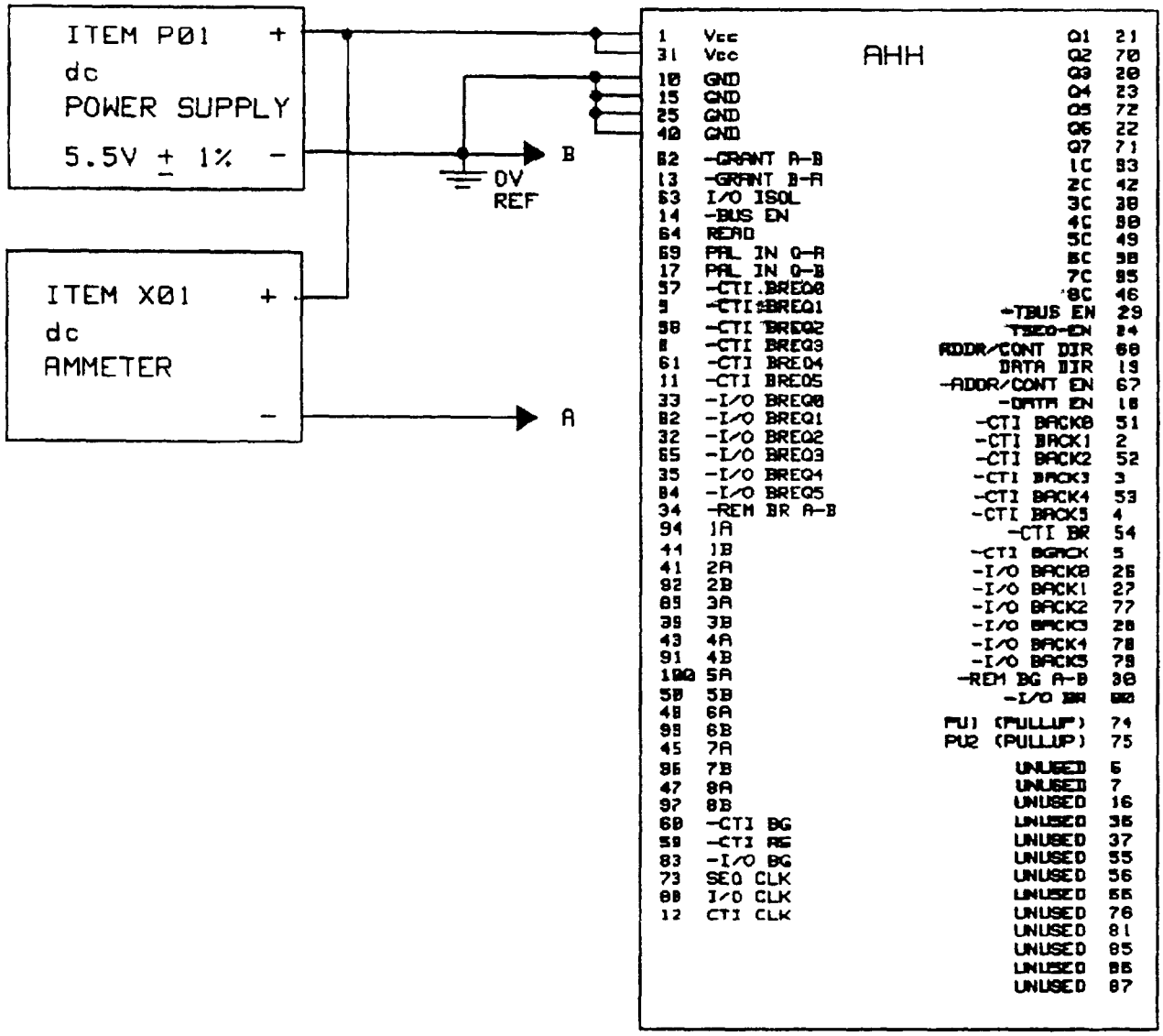
Test cond	Signal name	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
D1	-GRANT A-B	62	See note 2	0.0	1.0	mA	10
D2	-GRANT B-A	13					
D3	I/O ISOL	63					
D4	-BUS EN	14					
D5	READ	64					
D6	PAL IN Q-A	69					
D7	PAL IN Q-B	17	See note 2	0.0	1.0	mA	10
D8	-CTI BREQ0	57	See note 2	0.0	100.0	uA	10
D9	-CTI BREQ1	9					
D10	-CTI BREQ2	58					
D11	-CTI BREQ3	8					
D12	-CTI BREQ4	61					
D13	-CTI BREQ5	11					
D14	-I/O BREQ0	33					
D15	-I/O BREQ1	82					
D16	-I/O BREQ2	32					
D17	-I/O BREQ3	65					
D18	-I/O BREQ4	35					
D19	-I/O BREQ5	84					
D20	-REM BR A-B	34					
D21	1A	94	See note 2	0.0	100.0	uA	10

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
D22	1B	44	See note 2	0.0	100.0	uA	10
D23	2A	41					
D24	2B	92					
D25	3A	89					
D26	3B	39					
D27	4A	43					
D28	4B	91					
D29	5A	100					
D30	5B	50					
D31	6A	48					
D32	6B	99					
D33	7A	45					
D34	7B	96					
D35	8A	47					
D36	8B	97					
D37	-CTI BG	60					
D38	-CTI AS	59					
D39	-I/O BG	83					
D40	SEQ CLK	73	See note 2	0.0	100.0	uA	10
D41	I/O CLK	88	See note 2	0.0	1.10	mA	10
D42	CTI CLK	12	See note 2	0.0	1.10	mA	10

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NOTES:

1. Caution: Do not isolate the chassis of ammeter X01 from earth ground.

2. One input pin will be connected to point A, while all of the other input pins shall be connected to point B. Input pins are 62, 13, 63, 14, 64, 69, 17, 57, 9, 58, 8, 61, 11, 33, 82, 32, 65, 35, 84, 34, 94, 44, 41, 92, 89, 39, 43, 91, 100, 50, 48, 99, 45, 96, 47, 97, 60, 59, 83, 73, 88 and 12.

FIGURE 10. Input breakdown current test circuit for table V.

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TABLE VI. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Fig
E1	-GRANT A-B	62	-1.5	-0.4	V	11
E2	-GRANT B-A	13				
E3	I/O ISOL	63				
E4	-BUS EN	14				
E5	READ	64				
E6	PAL IN Q-A	69				
E7	PAL IN Q-B	17				
E8	-CTI BREQ0	57				
E9	-CTI BREQ1	9				
E10	-CTI BREQ2	58				
E11	-CTI BREQ3	8				
E12	-CTI BREQ4	61				
E13	-CTI BREQ5	11				
E14	-I/O BREQ0	33				
E15	-I/O BREQ1	82				
E16	-I/O BREQ2	32				
E17	-I/O BREQ3	65				
E18	-I/O BREQ4	35				
E19	-I/O BREQ5	84				
E20	-REM BR A-B	34	-1.5	-0.4	V	11

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TABLE VI. Input clamp diode voltage test - Continued.

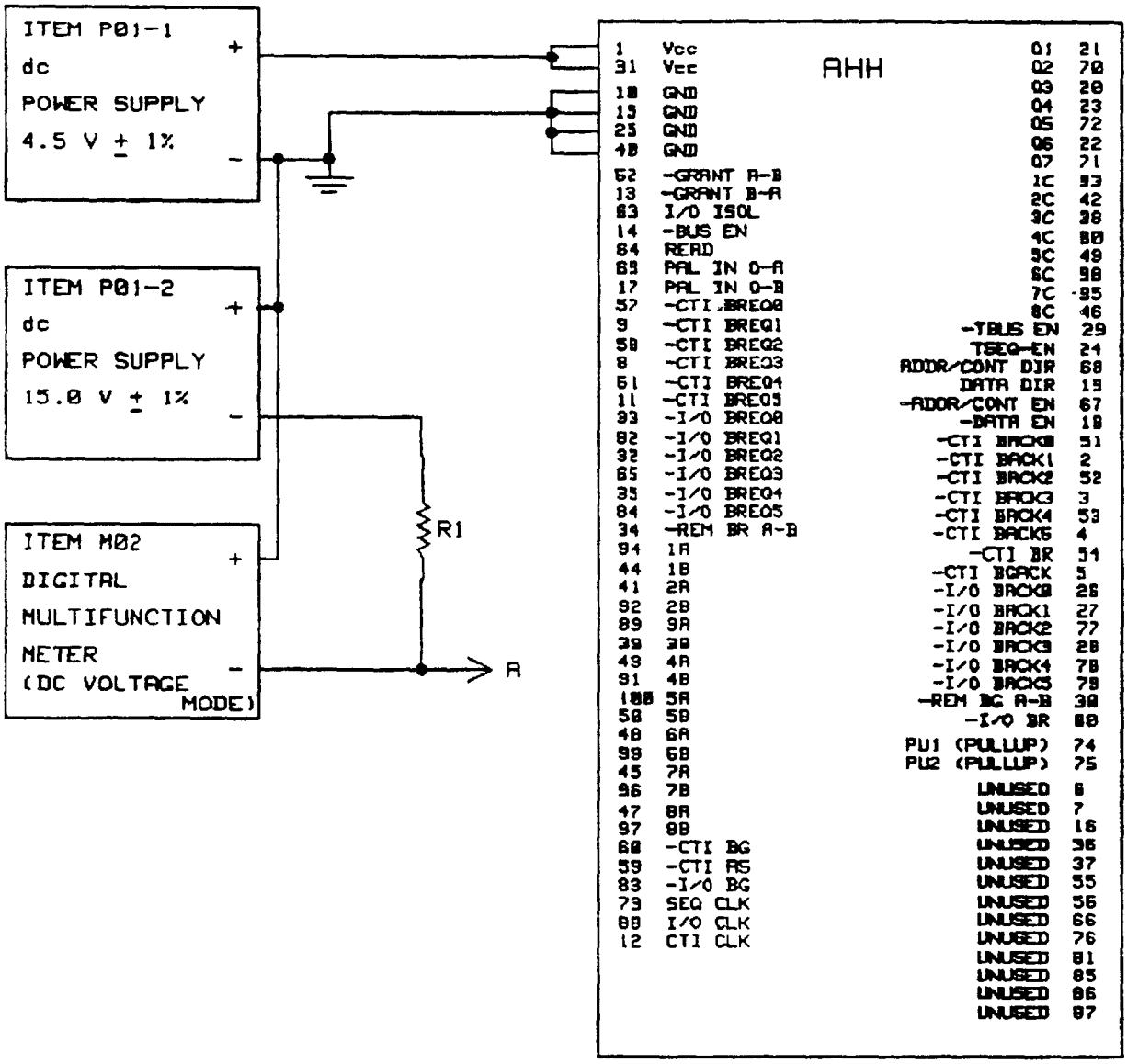
Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Fig
E21	1A	94	-1.5	-0.4	V	11
E22	1B	44				
E23	2A	41				
E24	2B	92				
E25	3A	89				
E26	3B	39				
E27	4A	43				
E28	4B	91				
E29	5A	100				
E30	5B	50				
E31	6A	48				
E32	6B	99				
E33	7A	45				
E34	7B	96				
E35	8A	47				
E36	8B	97				
E37	-CTI BG	60				
E38	-CTI AS	59				
E39	-I/O BG	83				
E40	SEQ CLK	73	-1.5	-0.4	V	11

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
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TABLE VI. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Fig
E41	I/O CLK	88	-1.5	-0.4	V	11
E42	CTI CLK	12	-1.5	-0.4	V	11

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NOTE:

1. R1 = 750 Ohm ± 1 percent, 1/4 W.

FIGURE 11. Input clamp diode voltage test circuit for table VI.

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TABLE VII. Output short circuit current test.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
F1	Q1	21	Open	73 3/	See note 5	-100.0	-15.0	mA	12
F2	Q2	70							
F3	Q3	20							
F4	Q4	23							
F5	Q5	72							
F6	Q6	22							
F7	Q7	71	Open	73 3/		-100.0	-15.0	mA	12
F8	1C	93	94,44	Open		-130.0	-15.0	mA	12
F9	2C	42	41,92						
F10	3C	38	89,39						
F11	4C	90	43,91			-130.0	-15.0	mA	12
F12	5C	49	100,50			-110.0	-15.0	mA	12
F13	6C	98	48,99						
F14	7C	95	45,96						
F15	8C	46	47,97	Open	See note 5	-110.0	-15.0	mA	12
F16	TSEQ-EN	24	Open	Open	See Note 5	-130.5	-30.1	mA	12
F17	ADDR/CONT DIR	68	Open	Open	See note 5	-130.0	-30.0	mA	12
F18	DATA DIR	19	Open	Open	See note 5	-130.0	-30.0	mA	12

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
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TABLE VII. Output short circuit current test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
F19	-ADDR/CONT EN	67	Open	Open	See note 5	-130.0	-30.0	mA	12
F20	-DATA EN	18	Open	Open	See note 5	-130.0	-30.0	mA	12
F21	-CTI BACK0	51	57,9,58,8, 61,11,60, 59	12 3/	See note 5	-250.0	-30.0	mA	12
F22	-CTI BACK1	2							
F23	-CTI BACK2	52							
F24	-CTI BACK3	3							
F25	-CTI BACK4	53							
F26	-CTI BACK5	4							
F27	-CTI BR	54							
F28	-CTI BGACK	5	57,9,58,8, 61,11,60, 59	12 3/					
F29	-I/O BACK0	26	33,82,32, 65,35,84, 34,83	88 3/					
F30	-I/O BACK1	27	33,82,32, 65,35,84, 34,83	88 3/	See Note 5	-250.0	-30.0	mA	12

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TABLE VII. Output short circuit current test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 + 5°C Initial limits Min	25 + 5°C Initial limits Max	Units	Figs
F31	-I/O BACK2	77	33,82,32, 65,35,84, 34,83	See Note 3	See Note 5	-250.0	-30.0	mA	12
F32	-I/O BACK3	28	"	"	"	"	"	"	"
F33	-I/O BACK4	78	"	"	"	"	"	"	"
F34	-I/O BACK5	79	"	"	"	"	"	"	"
F35	-REM BG A-B	30	"	"	"	"	"	"	"
F36	-I/O BR	80	33,82,32, 65,35,84, 34,83	See Note 3	See Note 5	-250.0	-30.0	mA	12

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Check the source to verify that this is the current version before use.

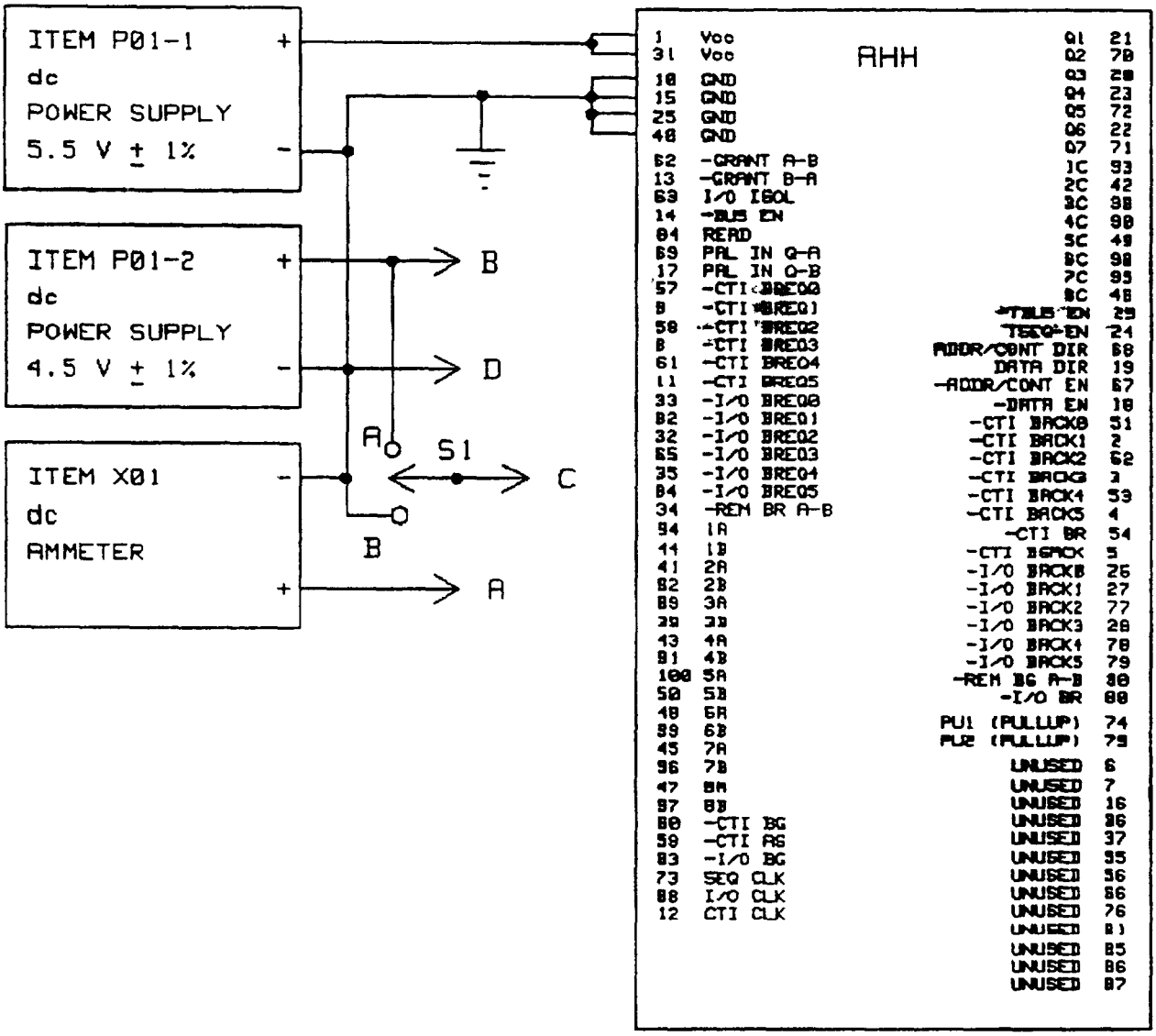


FIGURE 12. Output short circuit current test circuit for table VII.

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NOTES:

1. Do not isolate the chassis of ammeter X01 from earth ground.

2. S1 = DPST switch.

3. Alternate S1 from position A to position B, pausing a minimum of 100 nS after each move, until the desired output level has been achieved.

4. Only one output shall be shorted at a time and the measurement taken within 1 second.

5. Connect designated input pins to point B and C, while all of the other input pins shall be connected to point B. Input pins are 62,13,63,14,64,69,17,57,9,58,8,61,11,33,82,32,65,35 84,34,94,44,41,92,89,39,43,91,100,50,48,99,45,96,47,97,60,59, 83,73,88 and 12.

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TABLE VIII. Output voltage (low level) test.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
G1	Q1	21	Open	62,13	Open	Open	0.0	0.4	V	13
G2	Q2	70								
G3	Q3	20								
G4	Q4	23								
G5	Q5	72								
G6	Q6	22								
G7	Q7	71	Open	62,13						
G8	1C	93	94,44	Open						
G9	2C	42	41,92							
G10	3C	38	89,39							
G11	4C	90	43,91							
G12	5C	49	100,50							
G13	6C	98	48,99							
G14	7C	95	45,96							
G15	8C	46	47,97	Open	Open	Open	0.0	0.4	V	13
G16	-TBUS EN	29	Open	Open	Open	Open	0.0	0.5	V	13
G17	TSEQ-EN	24	Open	62,13						
G18	ADDR/CONT DIR	68	Open	13						
G19	DATA DIR	19	64	13	Open	Open	0.0	0.5	V	13

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Check the source to verify that this is the current version before use.

TABLE VIII. Output voltage (low level) test - Continued.

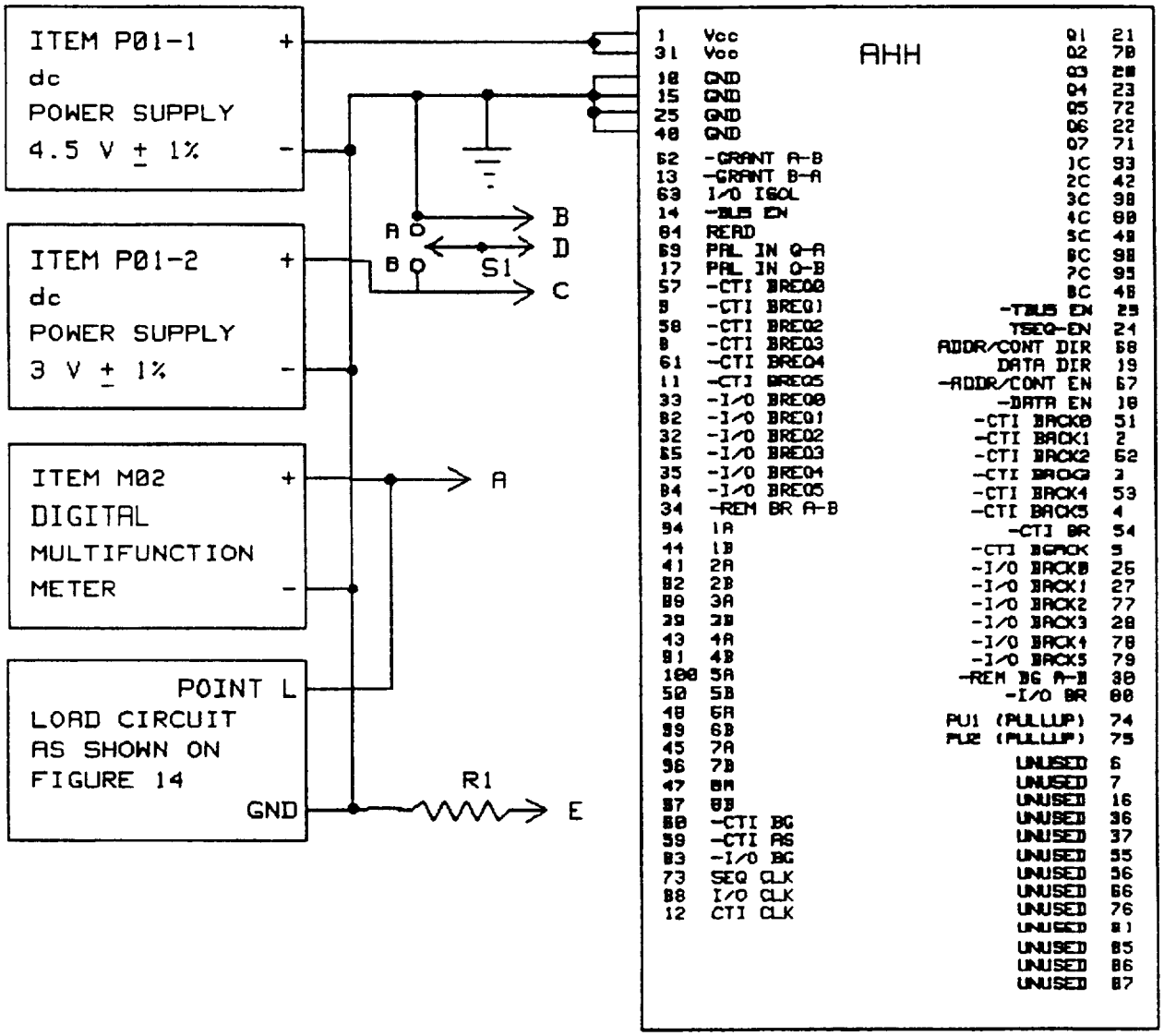
Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	25 + 5°C Initial limits Min	25 + 5°C Initial limits Max	Units	Fig
G20	-ADDR/CONT EN	67	13,63,14	69	Open	Open	0.0	0.5	V	13
G21	-DATA EN	18	13,63,14	17	Open	Open	0.0	0.5	V	13
G22	-CTI BACK0	51	57,60	9,58,8,61, 11,59	12 2/	Open	0.0	0.5	V	13
G23	-CTI BACK1	2	9,60	57,58,8,61, 11,59						
G24	-CTI BACK2	52	58,60	57,9,8,61, 11,59						
G25	-CTI BACK3	3	8,60	57,9,58,61, 11,59						
G26	-CTI BACK4	53	61,60	57,9,58,8, 11,59						
G27	-CTI BACK5	4	11,60	57,9,58,8, 61,59						
G28	-CTI BR	54	11	57,9,58,8, 59,60,61						
G29	-CTI BGACK	5	60	57,9,58,8, 11,59,61	12 2/	Open	0.0	0.5	V	13
G30	-I/O BACK0	26	33,83	82,32,65, 34,35,84	88 2/	Open	0.0	0.5	V	13
G31	-I/O BACK1	27	82,83	33,32,65, 34,35,84						
G32	-I/O BACK2	77	32,83	33,82,65, 34,35,84	88 2/	Open	0.0	0.5	V	13

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TABLE VIII. Output voltage (low level) test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	25 + 5°C Initial limits Min	25 + 5°C Initial limits Max	Units	Fig
G33	-I/O BACK3	28	65,83	33,82,32, 34,35,84	88 2/	Open	0.0	0.5	V	13
G34	-I/O BACK4	78	35,83	33,82,32, 65,84						
G35	-I/O BACK5	79	84,83	33,82,32, 34,65						
G36	-REM BG A-B	30	34,83	33,82,32, 33,65,84						
G37	-I/O BR	80	33	82,32,65, 34,35,83, 84	88 2/	Open	0.0	0.5	V	13

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

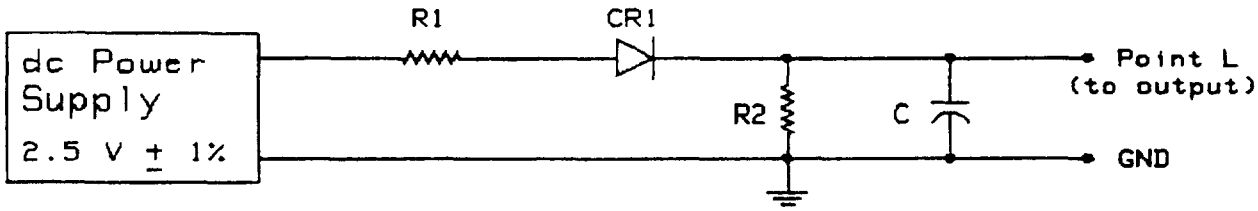


NOTES:

1. S1 = DPST switch.
2. Alternate S1 from position A to position B, pausing a minimum of 100 nS after each move, until the desired output level has been achieved.
3. R1 = 12 KOhms ± 1 percent, 1/8 W.

FIGURE 13. Output voltage (low-level and high-level) test circuit for tables VIII and IX.

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NOTES:

1. CR1 = 1N3600, 1N4150 or equivalent.

2. For output pins 20,21,22,23,38,42,46,49,70,71,72,90,93,95 and 98:
   R1 = Select value for a current flow of 4.0 mA ± 5 percent with point L held at 0.4V.
   R2 = 6.26 KOhm ± 1 percent, 1/8W.
   For output pin 24:
   R1 = Select value for a current flow of 7.56 mA ± 5 percent with point L held at 0.5V.
   R2 = 1.21 KOhm ± 1 percent, 1/8W.
   For output pins 18,19,67 and 68:
   R1 = Select value for a current flow of 8.0 mA ± 5 percent with point L held at 0.5V.
   R2 = 1.21 KOhm ± 1 percent, 1/8W.
   For output pin 29:
   R1 = Select value for a current flow of 7.5 mA ± 5 percent with point L held at 0.5V.

3. CL = 50 pf (including probe and parasitic capacitance) for crosstalk tests.
   CL = 150 pf (including probe and parasitic capacitance) for sample dynamic tests and propagation delay time.

4. Capacitor CL is not required for dc electrical test conditions G1-G37 and H1-H38.

5. Lead length from load circuit to module output pin under test is to be less than 1 inch (25.4 mm).

6. During output switching, the 2.5V power supply shall have a maximum overshoot of 500mV.

FIGURE 14. Logic load circuit.

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TABLE IX. Output voltage (high level) test.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
H1	Q1	21	62,13	Open	73 2/	Open	2.5	4.5	V	13
H2	Q2	70								
H3	Q3	20								
H4	Q4	23								
H5	Q5	72								
H6	Q6	22								
H7	Q7	71	62,13	Open	73 2/					
H8	1C	93	Open	94,44	Open					
H9	2C	42		41,92						
H10	3C	38		89,39						
H11	4C	90		43,91						
H12	5C	49		100,50						
H13	6C	98		48,99						
H14	7C	95		45,96						
H15	8C	46	Open	47,97	Open		2.5	4.5	V	13
H16	TSEQ-EN	24	62,13	Open	Open		2.4	4.5	V	13
H17	ADDR/CONT DIR	68	13	Open						
H18	DATA DIR	19	64,13	Open						
H19	-ADDR/CONT EN	67	69	63,13,62	Open	Open	2.4	4.5	V	13

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Check the source to verify that this is the current version before use.

TABLE IX. Output voltage (high level) test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	25 + 5°C Initial limits Min	25 + 5°C Initial limits Max	Units	Fig
H20	-DATA EN	18	17	63,13,62	Open	Open	2.4	4.5	V	13
H21	-CTI BACK0	51	Open	57,9,58,8,61, 11,60,59	12 2/	Open	2.4	4.5	V	13
H22	-CTI BACK1	2								
H23	-CTI BACK2	52								
H24	-CTI BACK3	3								
H25	-CTI BACK4	53								
H26	-CTI BACK5	4								
H27	-CTI BR	54								
H28	-CTI BGACK	5		57,9,58,8,61, 11,60,59	12 2/	Open	2.4	4.5	V	13
H29	-I/O BACK0	26		33,82,32,65, 35,84,34,83	88 2/	Open	2.4	4.5	V	13
H30	-I/O BACK1	27	Open	33,82,32,65, 35,84,34,83	88 2/					
H31	-I/O BACK2	77	Open		88 2/					
H32	-I/O BACK3	28								
H33	-I/O BACK4	78								
H34	-I/O BACK5	79								
H35	-REM BG A-B	30								
H36	-I/O BR	80	Open	33,82,32,65, 35,84,34,83	88 2/	Open	2.4	4.5	V	13

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE IX. Output voltage (high level) test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	25 + 5°C Initial T limits Min	25 + 5°C Initial T limits Max	Units	Fig
H37	PU1	74	Open	Open	Open	Open	2.4	2.6	V	13
H38	PU2	75	Open	Open	Open	Open	2.4	2.6	V	13

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE X. Input current (low level) test.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
J1	-GRANT A-B	62	Open	13,63,14,64,69,17	-786.8	-485.7	uA	15
J2	-GRANT B-A	13		62,63,14,64,69,17				
J3	I/O ISOL	63		62,13,14,64,69,17				
J4	-BUS EN	14		62,13,63,64,69,17	-786.8	-485.7	uA	15
J5	READ	64		62,13,63,14,69,17	-250.0	0.0	uA	15
J6	PAL IN Q-A	69		62,13,63,14,64,17				
J7	PAL IN Q-B	17		62,13,63,14,64,69	-250.0	0.0	uA	15
J8	-CTI BREQ0	57		9,58,8,61,11,60,59	-936.8	-585.7	uA	15
J9	-CTI BREQ1	9		57,58,8,61,11,60,59				
J10	-CTI BREQ2	58		57,9,8,61,11,60,59				
J11	-CTI BREQ3	8		57,9,58,61,11,60,59				
J12	-CTI BREQ4	61		57,9,58,8,11,60,59				
J13	-CTI BREQ5	11		57,9,58,8,61,60,59				
J14	-I/O BREQ0	33		82,32,65,35,84,34,83				
J15	-I/O BREQ1	82	Open	33,32,65,35,84,34,83	-936.8	-585.7	uA	15

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TABLE X. Input current (low level) test - Continued.

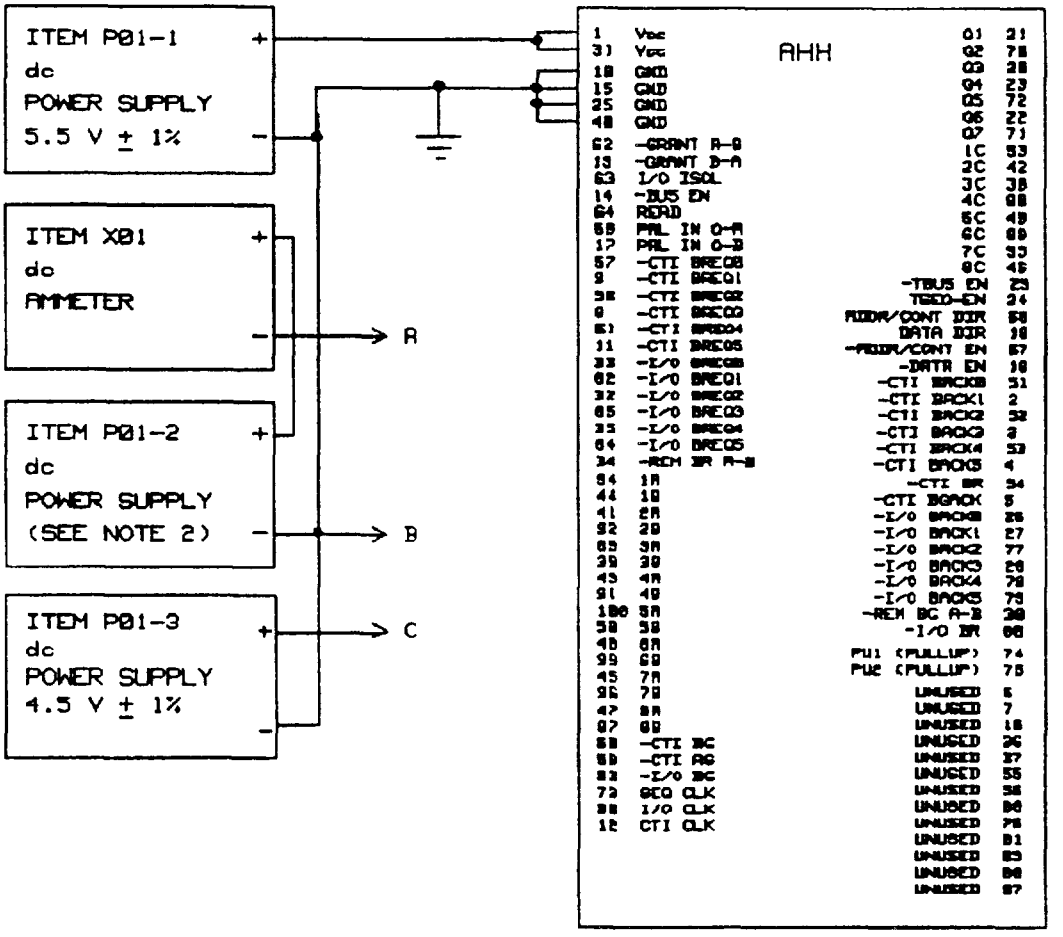
Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
J16	-I/O BREQ2	32	Open	32,82,65,35,84,34,83	-936.8	-585.7	uA	15
J17	-I/O BREQ3	65		33,82,32,35,84,34,83				
J18	-I/O BREQ4	35		33,82,32,65,84,34,83				
J19	-I/O BREQ5	84		33,83,32,65,35,34,83				
J20	-REM BR A-B	34		33,82,32,65,35,84,83	-936.8	-585.7	uA	15
J21	1A	94		44,41,92,89,39,43,91	-936.8	-515.7	uA	15
J22	1B	44		44,41,92,89,39,43,91				
J23	2A	41		94,49,92,89,39,43,91				
J24	2B	92		94,44,41,89,39,43,91				
J25	3A	89		94,44,41,92,39,43,91				
J26	3B	39		94,44,41,92,89,43,91				
J27	4A	43		94,44,41,92,89,39,91				
J28	4B	91		94,44,41,92,89,39,43				
J29	5A	100		50,48,99,45,96,47,97				
J30	5B	50	Open	100,48,99,45,96,47,97	-936.8	-515.7	uA	15

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TABLE X. Input current (low level) test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
J31	6A	48	Open	100,50,99,45,96,47,97	-936.8	-515.7	uA	15
J32	6B	99		100,50,48,45,96,47,97				
J33	7A	45		100,50,48,99,96,47,97				
J34	7B	96		100,50,48,99,45,47,97				
J35	8A	47		100,50,48,99,45,96,97				
J36	8B	97		100,50,48,99,45,96,47	-936.8	-515.7	uA	15
J37	-CTI BG	60		57,9,58,8,61,11,59	-400.0	-100.0	uA	15
J38	-CTI AS	59		57,9,58,8,61,11,60				
J39	-I/O BG	83		33,82,32,65,35,84,34	-400.0	-100.0	uA	15
J40	SEQ CLK	73		Open	-440.0	-100.0	uA	15
J41	I/O CLK	88		33,82,32,65,35,84,34,83	-650.0	-30.0	uA	15
J42	CTI CLK	12	Open	57,9,58,8,61,11,60,59	-650.0	-30.0	uA	15

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NOTES:

1. Caution: Do not isolate the chassis of ammeter X01 from earth ground.

2. For test conditions:
   J1-J42 - Item P01-2 = 0.4V ± 1 percent.
   K1-K7 - Item P01-2 = 2.4V ± 1 percent.
   K8- K42 - Item P01-2 = 2.7V ± 1 percent.

3. One input pin will be connected to point A, while all of the other input pins shall be connected to point B. Input pins are 62, 13, 63, 14, 64, 69, 17, 57, 9, 58, 8, 61, 11, 33, 82, 32, 65, 35, 84, 34, 94, 44, 41, 92, 89, 39, 43, 91, 100, 50, 48, 99, 45, 96, 47, 97, 60, 59, 83, 73, 88 and 12.

FIGURE 15. Input current (low-level and high-level) test circuit for tables X and XI.

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TABLE XI. Input current (high level) test.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
K1	-GRANT A-B	62	See note 3 of figure 15	Open	-351.3	-295.2	uA	15
K2	-GRANT B-A	13						
K3	I/O ISOL	63						
K4	-BUS EN	14			-351.3	-295.2	uA	15
K5	READ	64			0.0	25.0	uA	15
K6	PAL IN Q-A	69						
K7	PAL IN Q-B	17			0.0	25.0	uA	15
K8	-CTI BREQ0	57			-314.7	-266.7	uA	15
K9	-CTI BREQ1	9						
K10	-CTI BREQ2	58						
K11	-CTI BREQ3	8						
K12	-CTI BREQ4	61						
K13	-CTI BREQ5	11						
K14	-I/O BREQ0	33						
K15	-I/O BREQ1	82	See note 3 of figure 15	Open	-314.7	-266.7	uA	15

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TABLE XI. Input current (high level) test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
K16	-I/O BREQ2	32	See note 3 of figure 15	Open	-314.7	-266.7	uA	15
K17	-I/O BREQ3	65						
K18	-I/O BREQ4	35						
K19	-I/O BREQ5	84						
K20	-REM BR A-B	34						
K21	1A	94						
K22	1B	44						
K23	2A	41						
K24	2B	92						
K25	3A	89						
K26	3B	39						
K27	4A	43						
K28	4B	91						
K29	5A	100						
K30	5B	50	See note 3 of figure 15	Open	-314.7	-266.7	uA	15

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TABLE XI. Input current (high level) test - Continued.

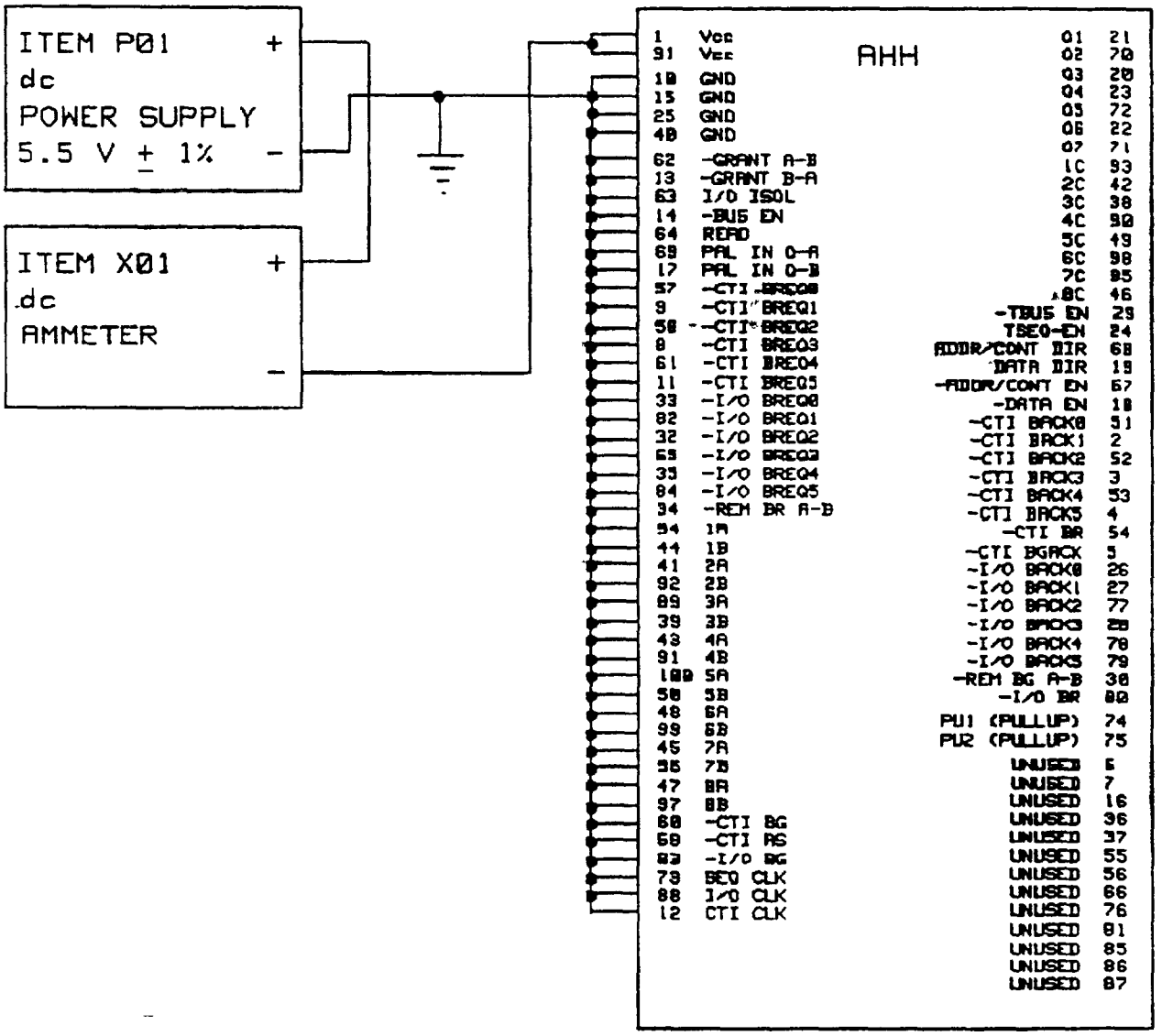
Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 + 5°C Initial limits Min	25 + 5°C Initial limits Max	Units	Figs
K31	6A	48	See note 3 of figure 15	Open	-314.7	-266.7	uA	15
K32	6B	99						
K33	7A	45						
K34	7B	96						
K35	8A	47						
K36	8B	97			-314.7	-266.7	uA	15
K37	-CTI BG	60			0.0	20.0	uA	15
K38	-CTI AS	59						
K39	-I/O BG	83						
K40	SEQ CLK	73			0.0	20.0	uA	15
K41	I/O CLK	88			0.0	45.0	uA	15
K42	CTI CLK	12	See note 3 of figure 15	Open	0.0	45.0	uA	15

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TABLE XII. Power supply current test.

Test cond	Parameter	25 ± 5°C Initial limits Min	Max	Units	Fig.
L	Icc	57.3	572.7	mA	16

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NOTE:

Caution: Do not isolate the chassis of ammeter X01 from earth ground.

FIGURE 16. Power supply current test circuit for table XII.

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TABLE XIII. Propagation delay time (tPLH) test.

Test cond	Signal name	Connect A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	25 ± 5°C Initial T limits Min	Max	Units	Figs
M1	Q1	21	73	Open	71	13	Open	62	5	44	nS	17, 18
M2	Q2	78										
M3	Q3	28										
M4	Q4	23										
M5	Q5	72										
M6	Q6	22										
M7	Q7	71	73	Open	71	13	Open	62	5	44	nS	17, 18
M8	1C	93	94	Open	Open	Open	Open	44	2	27	nS	17, 18
M9	1C	93	44					94				
M10	2C	42	41					92				
M11	2C	42	92					41				
M12	3C	38	89					39				
M13	3C	38	39					89				
M14	4C	98	43					91				
M15	4C	98	91	Open	Open	Open	Open	43	2	27	nS	17, 18
M16	5C	48	100	Open	Open	Open	50	Open	2	32	nS	17, 18
M17	5C	49	50				100					
M18	6C	98	48				99					
M19	6C	98	99				48					
M20	7C	95	45				96					
M21	7C	95	95				45					
M22	8C	48	47				97					
M23	8C	48	97	Open	Open	Open	47	Open	2	32	nS	17, 18

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TABLE XIII. Propagation delay time (tPLH) test - Continued.

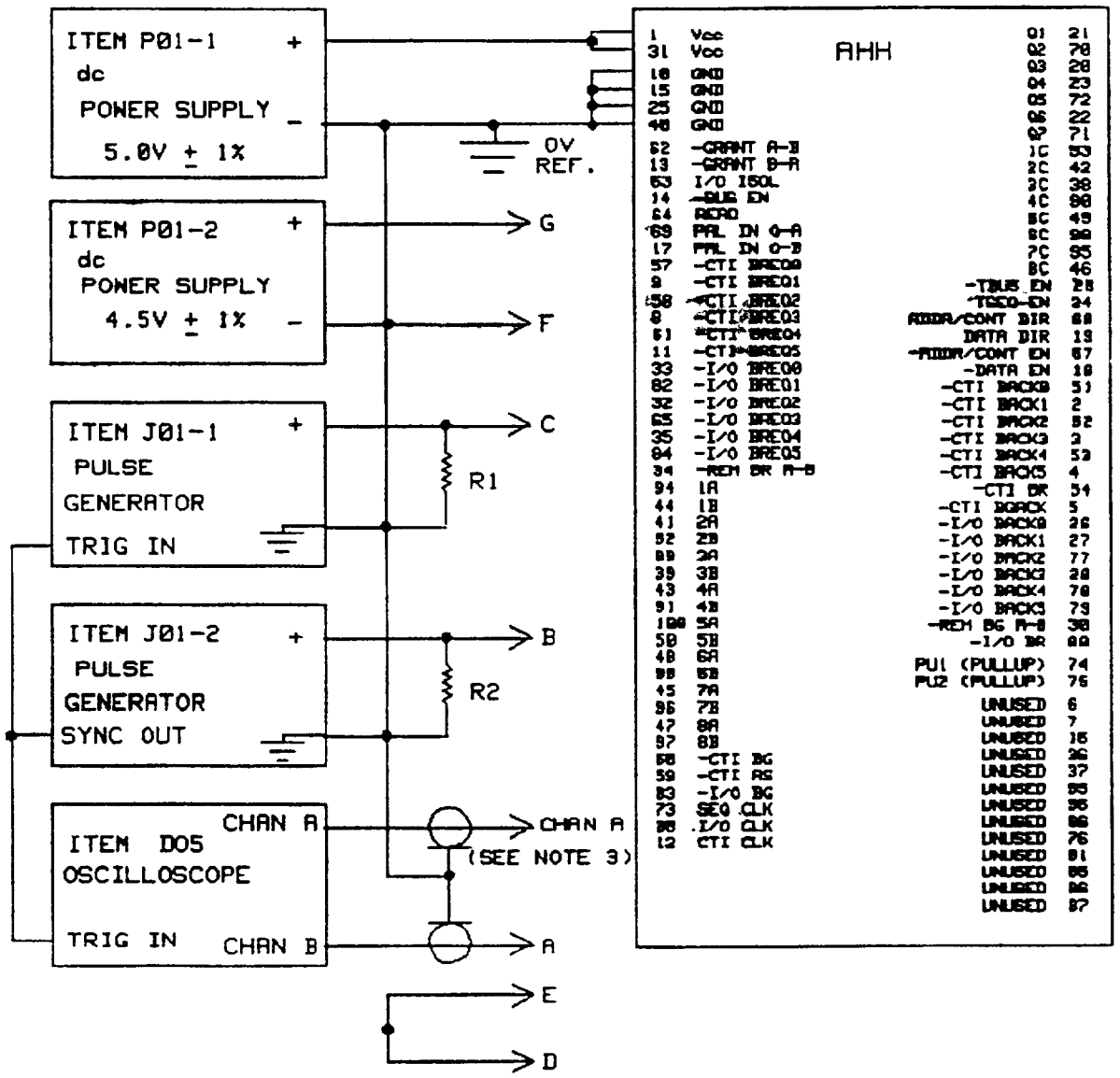
Test cond	Signal name	Connect A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
M24	TSEQ-EN	24	13	Open	Open	Open	Open	62	5	52	nS	17,18
M25	TSEQ-EN	24	62				Open	13				
M26	ADDR/CONT DIR	68	13				Open	Open				
M27	DATA DIR	19	13				64	Open				
M28	DATA DIR	19	13				Open	64				
M29	DATA DIR	19	64				13	Open				
M30	DATA DIR	19	64	Open	Open	Open	Open	13	5	52	nS	17,18
M31	-ADDR/CONT EN	67	13	Open	Open	Open	63,14	62,69	5	52	nS	17,18
M32	-ADDR/CONT EN	67	62				63,14	13,69				
M33	-ADDR/CONT EN	67	63				13,14	69,62				
M34	-ADDR/CONT EN	67	63				62,14	69,13				
M35	-ADDR/CONT EN	67	14				13,63	69,62				
M36	-ADDR/CONT EN	67	14				62,63	13,69				
M37	-ADDR/CONT EN	67	69				13,63,14	62				
M38	-ADDR/CONT EN	67	69				62,63,14	13				
M39	-DATA EN	18	13				63,14	17,62				
M40	-DATA EN	18	62				63,14	13,17				
M41	-DATA EN	18	63				13,14	17,62				
M42	-DATA EN	18	63				62,14	17,13				
M43	-DATA EN	18	14				13,63	17,62				
M44	-DATA EN	18	14				62,63	13,17				
M45	-DATA EN	18	17	Open	Open	Open	13,14,63	62	5	52	nS	17,18

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TABLE XIII. Propagation delay time (tPLH) test - Continued.

Test cond	Signal name	Connect A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	25 + 5°C Initial T limits Min	Max	Units	Figs
M46	-DATA EN	10	17	Open	Open	Open	62,63,14	13	5	52	nS	17,18
M47	-CTI BACK0	51	12	57,60			Open	50,9,59,8,61,11	0	27	nS	17,18
M48	-CTI BACK1	2		9,60				57,59,8,61,11,59				
M49	-CTI BACK2	52		58,60				57,9,8,61,11,58				
M50	-CTI BACK3	3		8,60				57,9,58,61,11,58				
M51	-CTI BACK4	53		61,60				57,9,58,8,11,58				
M52	-CTI BACK5	4		11,60				57,9,58,8,61,59				
M53	-CTI BR	54	12	11				57,9,58,8,61,60,5				
M54	-CTI BGACK	5	88	60				57,9,58,8,61,11,5				
M55	-I/O BACK0	26		33,83				82,32,65,35,84,34				
M56	-I/O BACK1	27		82,83				33,32,65,35,84,34				
M57	-I/O BACK2	77		32,83				33,82,65,35,84,34				
M58	-I/O BACK3	28		65,83				33,82,32,35,84,34				
M59	-I/O BACK4	78		35,83				33,82,32,65,84,34				
M60	-I/O BACK5	79	88	84,83	Open	Open	Open	33,82,32,65,35,34				
M61	-REM BG A-B	30	88	34,83	Open	Open	Open	33,82,32,65, 35,84				
M62	-I/O BR	80	88	33	Open	Open	Open	82,32,65,35, 84,34,83	0	27	nS	17,18

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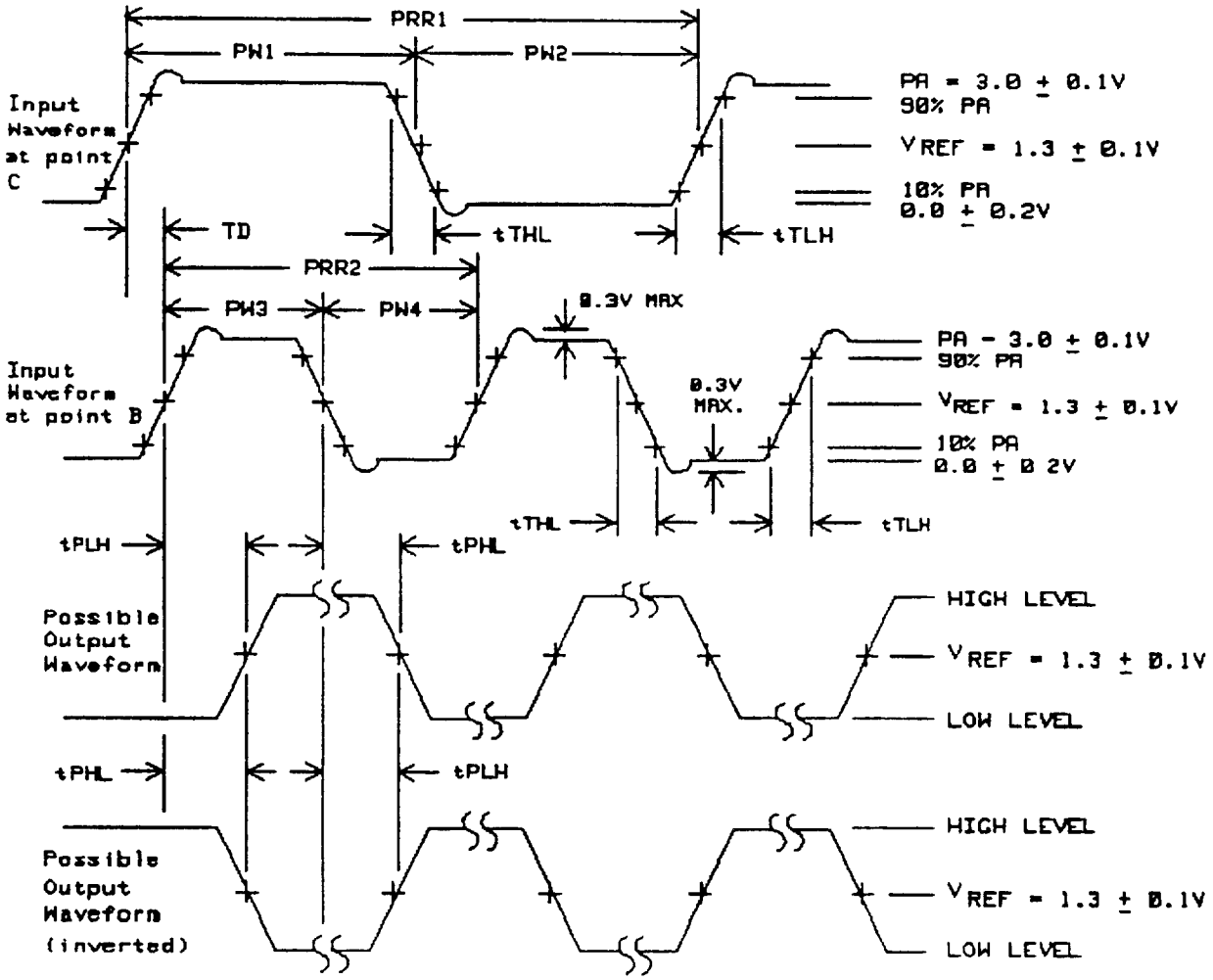


NOTES:

1. R1 = R2 = 51.1 Ohm ± 5 percent.
2. Load each output per figure 14.
3. For test conditions M1-M62, N8-N62, connect CHAN A to point B. For test conditions N1-N7, connect CHAN A to module pin 24.

FIGURE 17. Propagation delay time (tPLH and tPHL) test circuit for tables XIII and XIV.

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NOTES:

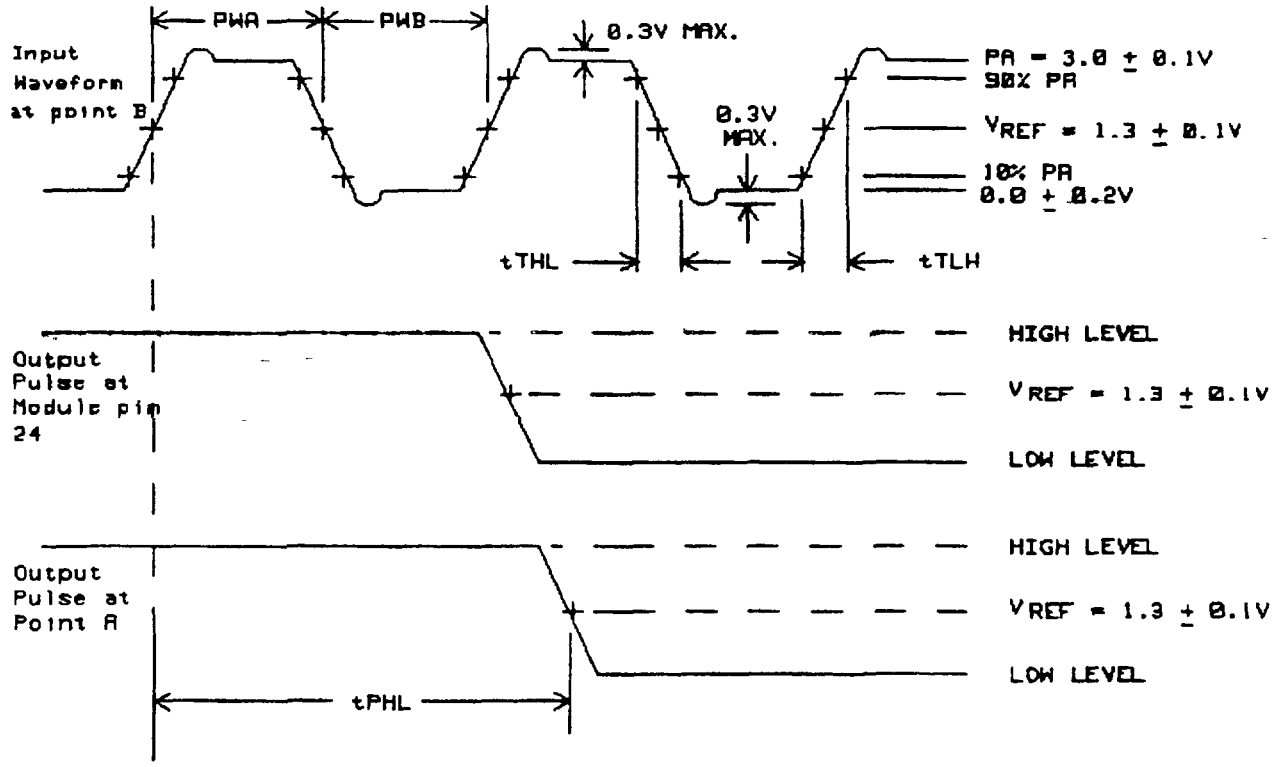
1. Input pulse parameters:

   PW1 = PW2 = 500 nS ± 10 percent
   PW3 = PW4 = 250 nS ± 10 percent
   PRR2 = 2 mHZ ± 10 percent
   PRR1 = 1 mHZ ± 10 percent
   tTHL = tTLH = 10 nS ± 2 nS
   TD = 30 nS ± 2 nS

2. Before making the measurements, verify that the output pulse high level is ≥ 2.5 V and low level is ≤ 0.4 V.

FIGURE 18. Propagation delay time (tPLH and tPHL) test circuit waveforms for tables XIII and XIV.

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NOTES:

1. Input pulse parameters:

   PWA = PWB = 250 nS ± 10 percent
   PRR = 2 mHZ ± 10 percent
   tTLH = tTHL = 10 ± 2 nS

2. Before making the measurements, verify that the output pulse high level is ≥ 2.5V and the low level is ≤ 0.4V.

FIGURE 19. Propagation delay time (tPLH and tPHL) test circuit waveforms for tables XIII and XIV.

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TABLE XIV. Propagation delay time (tPHL) test.

Test cond	Signal name	Connect A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	25 ± 5°C Initial limits Min	Max	Units	Figs
N1	Q1	21	73	Open	71	13	Open	62	5	52	nS	17,19
N2	Q2	70										
N3	Q3	20										
N4	Q4	23										
N5	Q5	72										
N6	Q6	22										
N7	Q7	71	73		71	13		62	5	52	nS	17,19
N8	1C	93	94		Open	Open		44	2	31	nS	17,18
N9	1C	93	44					94				
N10	2C	42	41					92				
N11	2C	42	92					41				
N12	3C	38	89					39				
N13	3C	38	39					89				
N14	4C	90	43					91				
N15	4C	90	91	Open	Open	Open	Open	43	2	31	nS	17,18
N16	5C	49	100	Open	Open	Open	50	Open	2	31	nS	17,18
N17	5C	49	50				100					
N18	6C	98	48				99					
N19	6C	98	99				48					
N20	7C	95	45				96					
N21	7C	95	96	Open	Open	Open	45	Open	2	31	nS	17,18

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TABLE XIV. Propagation delay time (tPHL) test - Continued.

Test cond	Signal name	Connect A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	25 ± 5°C Initial limits Min	Max	Units	Figs
N22	BC	46	47	Open	Open	Open	97	Open	2	31	nS	17,18
N23	BC	46	97				47	Open	2	31	nS	17,18
N24	TSEQ-EN	24	13				Open	62	5	56	nS	17,18
N25	TSEQ-EN	24	62				Open	13				
N26	ADDR/CONT DIR	88	13				Open	Open				
N27	DATA DIR	18	13				64	Open				
N28	DATA DIR	18	13				Open	64				
N29	DATA DIR	19	64				13	Open				
N30	DATA DIR	19	64	Open	Open	Open	Open	13	5	56	nS	17,18
N31	-ADDR/CONT EN	67	13	Open	Open	Open	63,14	62,69	5	56	nS	17,18
N32	-ADDR/CONT EN	67	62				63,14	13,69				
N33	-ADDR/CONT EN	67	63				13,14	69,62				
N34	-ADDR/CONT EN	67	63				62,14	69,13				
N35	-ADDR/CONT EN	67	14				13,63	69,62				
N36	-ADDR/CONT EN	67	14				62,63	13,69				
N37	-ADDR/CONT EN	67	69				13,63,14	62				
N38	-ADDR/CONT EN	67	69				62,63,14	13				
N39	-DATA EN	18	13				63,14	17,62				
N40	-DATA EN	18	62				63,14	13,17				
N41	-DATA EN	18	63				13,14	17,62				
N42	-DATA EN	18	63				62,14	17,13				
N43	-DATA EN	18	14	Open	Open	Open	13,63	17,62	5	56	nS	17,18

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TABLE XIV. Propagation delay time (tPHL) - Continued.

Test cond	Signal name	Connect A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	25 + 5°C Initial limits Min	Max	Units	Figs
N44	-DATA EN	18	14	Open	Open	Open	62,63	13,17	5	58	nS	17,18
N45	-DATA EN	18	17	Open	Open	Open	13,63,14	62	5	58	nS	17,18
N46	-DATA EN	18	17	Open	Open	Open	62,63,14	13	5	58	nS	17,18
N47	-CTI BACK0	51	12	57,60			Open	58,9,59, 11,61,8	0	28	nS	17,18
N48	-CTI BACK1	2		9,60				57,58,8, 11,59,61				
N49	-CTI BACK2	52		58,60				57,61,59, 9,8,11				
N50	-CTI BACK3	3		8,60				57,58,61, 9,11,59				
N51	-CTI BACK4	53		61,60				57,58,8, 9,11,59				
N52	-CTI BACK5	4		11,60				57,9,58, 8,59,61				
N53	-CTI BR	54		11				57,58,61, 8,9,5,60				
N54	-CTI BGACK	5	12	60				57,58,61, 8,9,11,59				
N55	-I/O BACK0	28	88	33,83				82,32,65, 34,35,84				
N56	-I/O BACK1	27		82,83				33,32,65, 34,35,84				
N57	-I/O BACK2	77		32,83				33,82,65, 34,35,84				
N58	-I/O BACK3	28		65,83				33,82,32, 34,35,84				
N59	-I/O BACK4	78		35,83				33,82,32, 34,65,84				
N60	-I/O BACK5	79	88	84,83	Open	Open	Open	33,82,32, 34,35,65	0	28	nS	17,18

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TABLE XIV. Propagation delay time (tPHL) test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	Connect F to pin	Connect G to pin	25 ± 5°C Initial T limits Min	25 ± 5°C Initial T limits Max	Units	Figs
N61	-REM BG A-B	30	88	34,83	Open	Open	Open	33,82,32, 35,65,84	0	28	nS	17,18
N62	-I/O BR	80	88	33	Open	Open	Open	82,32,65, 34,35,83, 84	0	28	nS	17,18

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TABLE XV. Dynamic tests.

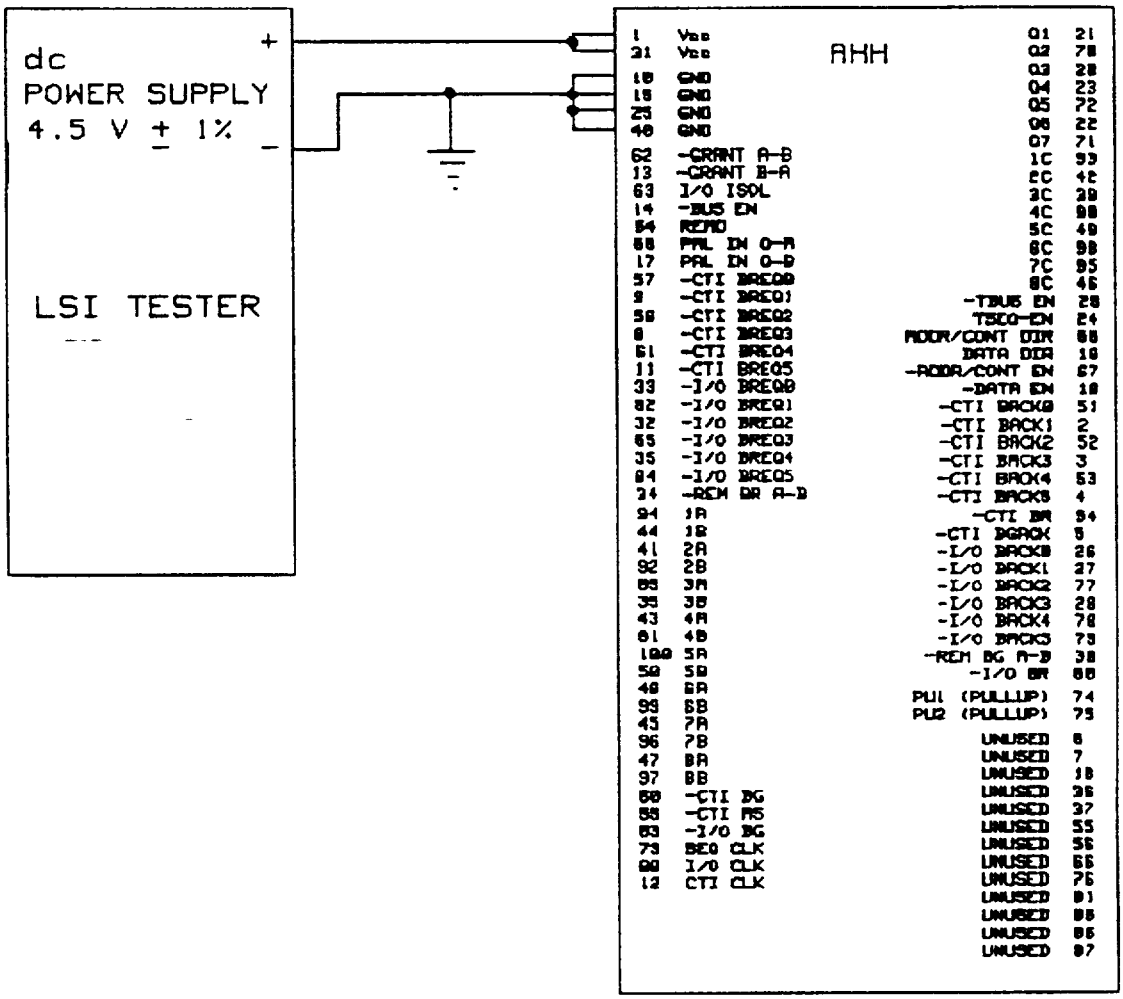
Test cond	Parameter	Connect	Initial limits 25°± 5°C		Test figures
			Min.	Max.	
P1-P16	-CTI BREQ 0-5, -I/O BREQ 0-5, -CTI BG, -CTI AS, -I/O BG, -REM BR A-B setup time (tSU)	See the appendix for test vectors	Module shall pass functional test defined in table XVI.		20,21
Q1-Q16	-CTI BREQ 0-5, -I/O BREQ 0-5, -CTI BG, -CTI AS, -I/O BG, -REM BR A-B hold time (tH)				20,21
R1-R3	CTI CLK, I/O CLK, SEQ CLK MAX clock Frequency (F Max)				20,22
R4-R6	CTI CLK, I/O CLK, SEQ CLK MIN pulse width (PW)				20,23
R7	TSEQ-EN MIN Pulse width (PW)	See the appendix for test vectors	Module shall pass functional test defined in table XVI.		

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TABLE XVI. Functional test.

Test cond	Parameter	Connect	Initial limits 25°± 5°C		Test figure
			Min.	Max.	
S	Functional test	See the appendix for test vectors	Module shall pass test		20

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NOTES:

1. See the appendix for test vectors.

2. Input conditioning voltages will be 0.4V ± 1 percent for a logic 0 and 2.4V ± 1 percent for a logic 1.

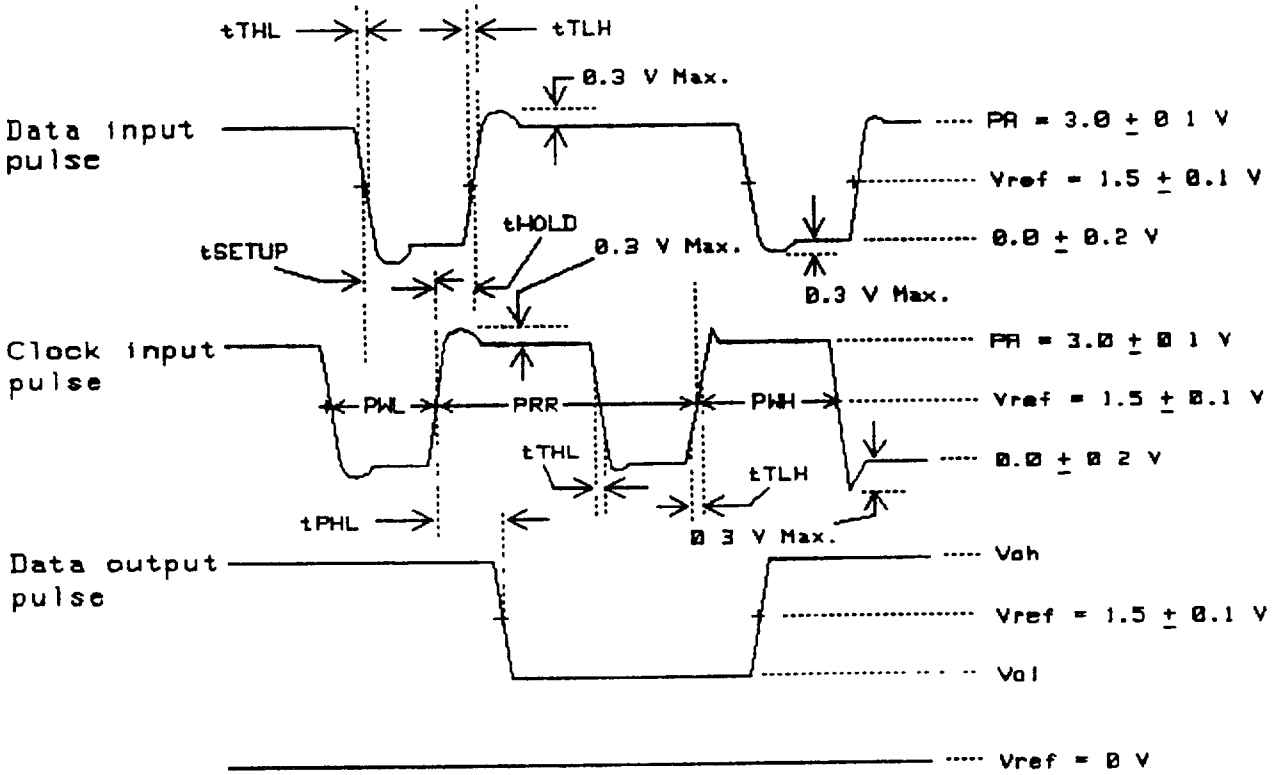
3. Load all outputs per figure 14 for test conditions: P1-P16, Q1-Q16 and R1-R7.

4. Logic levels in the appendix:

   H = VOH (table IX)
   L = VOL (table VIII)

FIGURE 20. Functional test circuit for tables XV and XVI.

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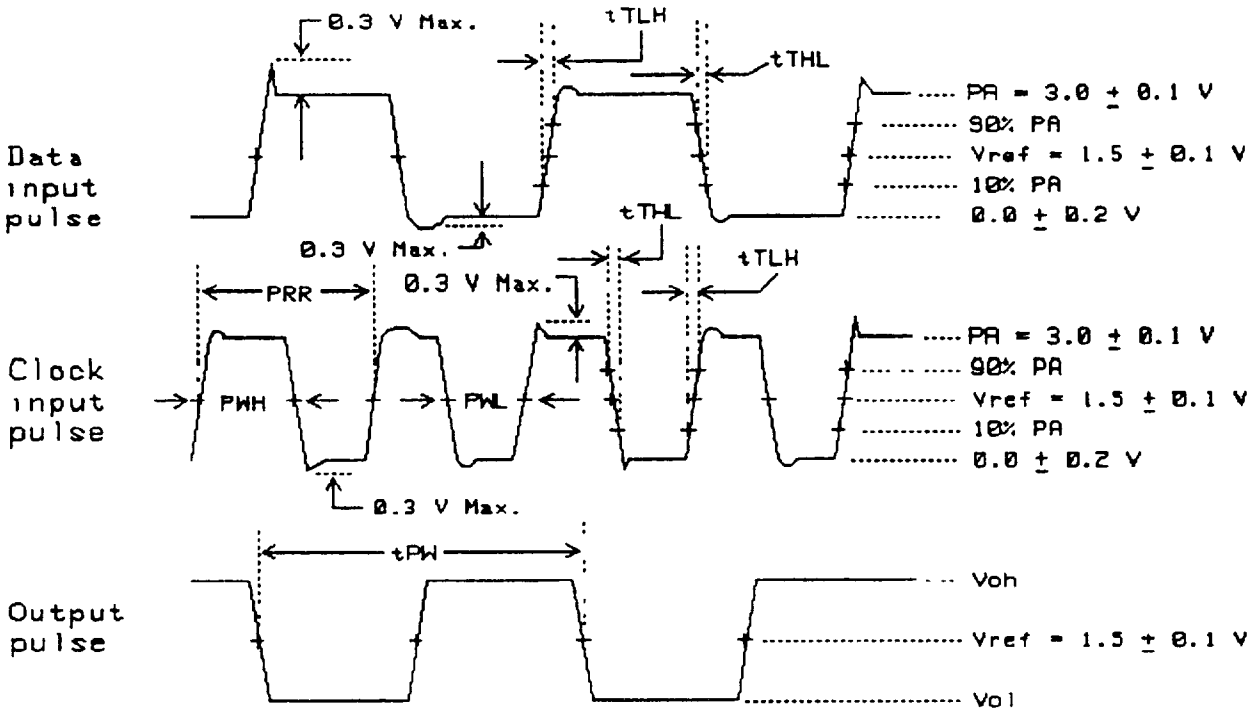
NOTES:

1. Data input pulse parameters:
   tTHL = 6 nS ± 1.5 nS
   tTLH = 15 nS ± 1.5 nS
   tSETUP = 20 nS ± 0.5 nS
   tHOLD = 5 nS ± 0.5 nS

2. Clock input pulse parameters:
   tTHL = 6 nS ± 1.5 nS
   tTLH = 15 nS ± 1.5 nS
   PRR = 500 nS min.
   PWH = 250 nS min.
   PWL = 250 nS min.

FIGURE 21. Setup (tSU) and hold time (tH) test circuit waveforms for tables XV and XXII.

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NOTES:

1. Data input pulse parameters:

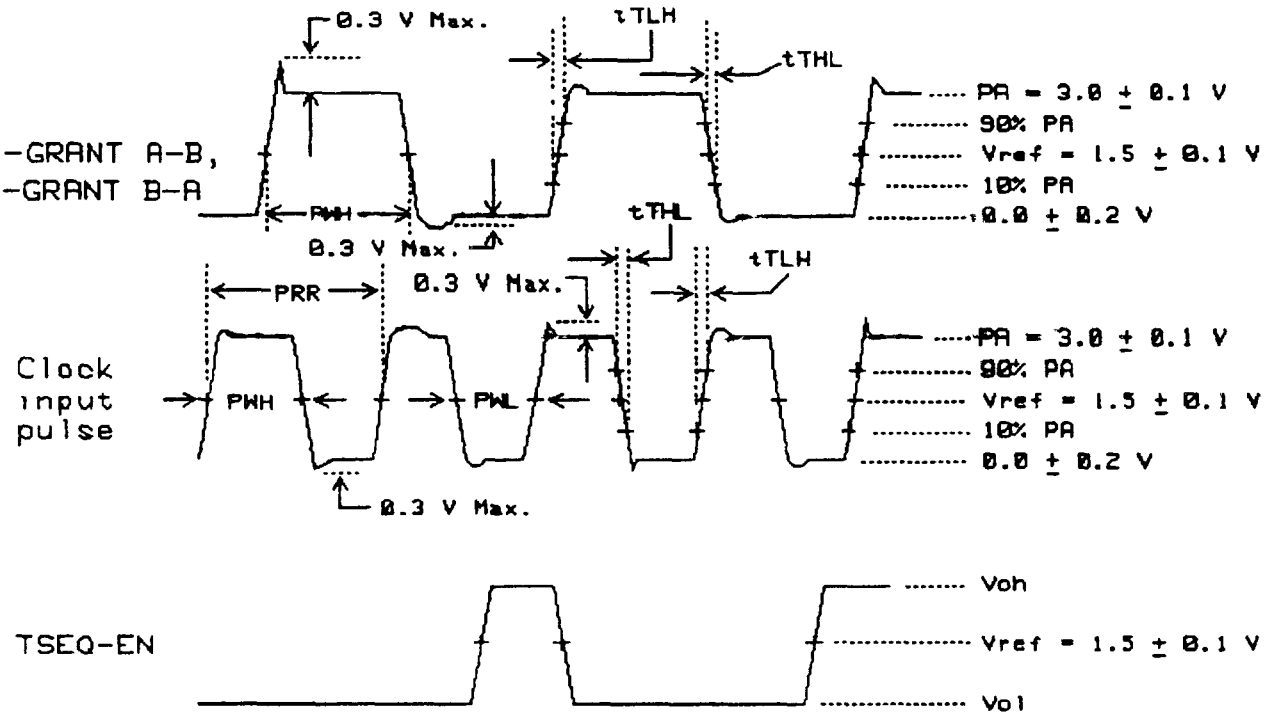
   tTHL = 6 nS ± 1.5 nS
   tTLH = 15 nS ± 1.5 nS

2. Clock input pulse parameters:

   tTHL = 6 nS ± 1.5 nS
   tTLH = 15 nS ± 1.5 nS
   PRR = 100 nS ± 10 nS
   PWH = 20 nS ± 2 nS
   PWL = 80 ± 5 nS

FIGURE 22. Maximum clock frequency, minimum pulse width high and minimum pulse width low test waveforms for table XV.

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NOTES:

1. Apply a minimum of 8 clock pulses to pin 73 (SEQ CLK) before applying a high level to pin 62 (-GRANT A-B) and pin 13 (-GRANT B-A) to clear the shift register.

2. -GRANT A-B and -GRANT B-A input pulse parameters:
   PWH = 70 nS min. ± 5 nS.

3. Clock input pulse parameters:

   tTHL = 6 nS ± 1.5 nS, tTLH = 15 nS ± 1.5 nS
   PRR = 500 nS ± 10 nS, PWL = PWH = 250 nS ± 5 nS

FIGURE 23. Dynamic waveforms for table XV.

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TABLE XVII. Output crosstalk, low level, test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs
T1	Q1	21	0	400	mV	1-4	24,25
T2	Q2	70					
T3	Q3	20					
T4	Q4	23					
T5	Q5	72					
T6	Q6	22					
T7	Q7	71					
T8	ADDR/CONT DIR	68					
T9	DATA DIR	19					
T10	-ADDR/CONT EN	67					
T11	-DATA EN	18					
T12	-CTI BACK0	51					
T13	-CTI BACK1	2					
T14	-CTI BACK2	52					
T15	-CTI BACK3	3					
T16	-CTI BACK4	53					
T17	-CTI BACK5	4					
T18	-CTI BR	54					
T19	-CTI BGACK	5					
T20	-I/O BACK0	26	0	400	mV	1-4	24,25

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TABLE XVII. Output crosstalk, low level, test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs
T21	-I/O BACK1	27	0	400	mV	1-4	24,25
T22	-I/O BACK2	77					
T23	-I/O BACK3	28					
T24	-I/O BACK4	78					
T25	-I/O BACK5	79					
T26	-REM BG A-B	30					
T27	-I/O BR	80	0	400	mV	1-4	24,25
T28	1C	93	0	500	mV	1-4	24,25
T29	2C	42					
T30	3C	38					
T31	4C	90					
T32	5C	49					
T33	6C	98					
T34	7C	95					
T35	8C	46	0	500	mV	1-4	24,25

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
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NOTES:

1. Output crosstalk, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 0.25 V	T1-T35
Power supply current limit	0.724 A	NA	T1-T35
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 T1-T35 T1-T35
Output crosstalk measurement	 NA NA	 ± 20 mV ± 25 mV	 T1-T27 T28-T35

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) Use probe 'A' to verify the pattern voltage and pattern repetition rate of 1000 nS ± 200 nS, 50 percent duty cycle.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
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NOTES: - Continued.

(d) For each output under test in table XVII, connect probe 'B' and visually inspect the output for the most positive spike as indicated on figure 25.

(e) The maximum probe capacitance is 10 pF.

3. All outputs shall be loaded as specified on figure 14.

4. Abbreviations used are defined in 6.1.

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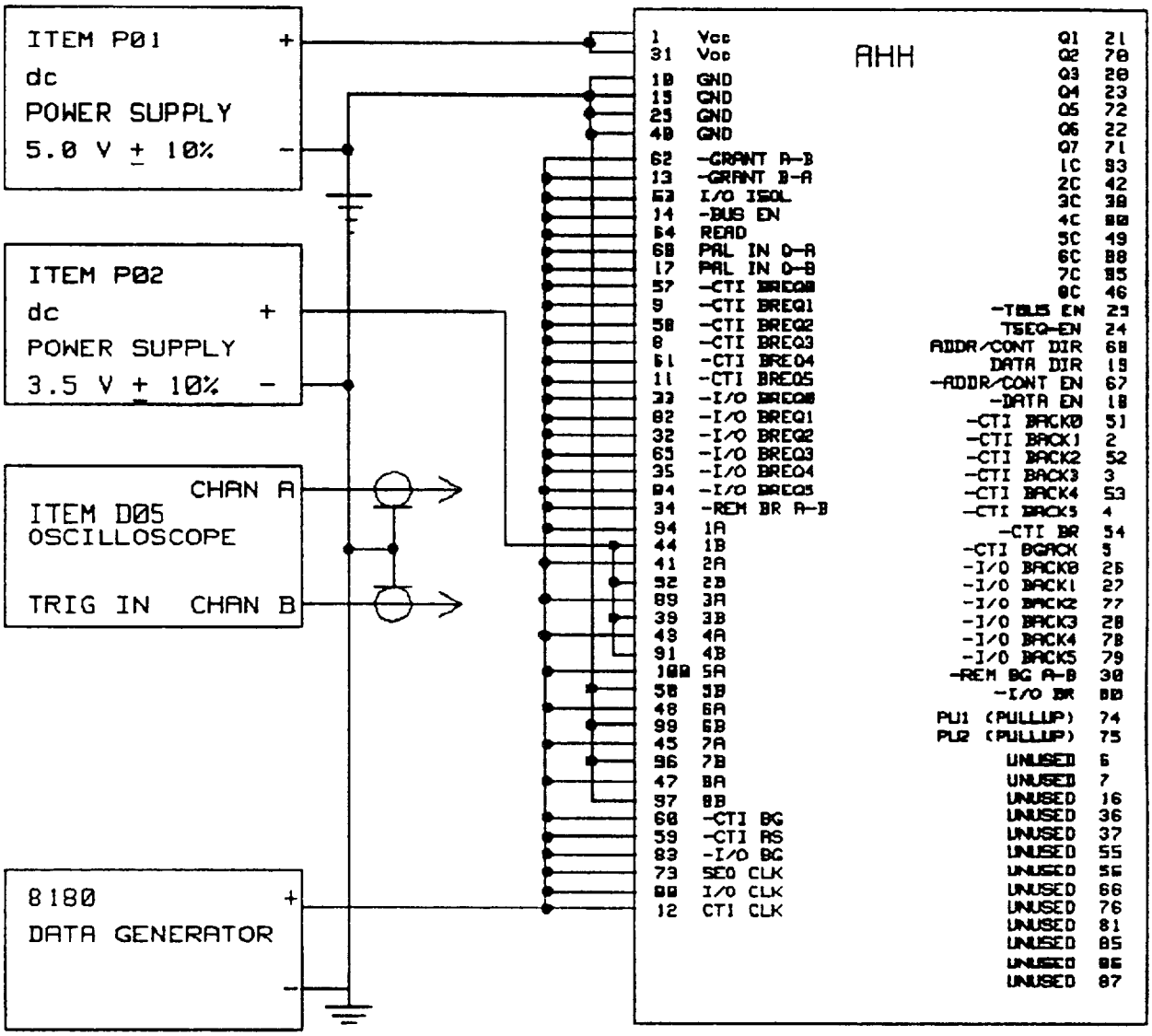


FIGURE 24. Output crosstalk, low level and high level, test circuit for tables XVII and XVIII.

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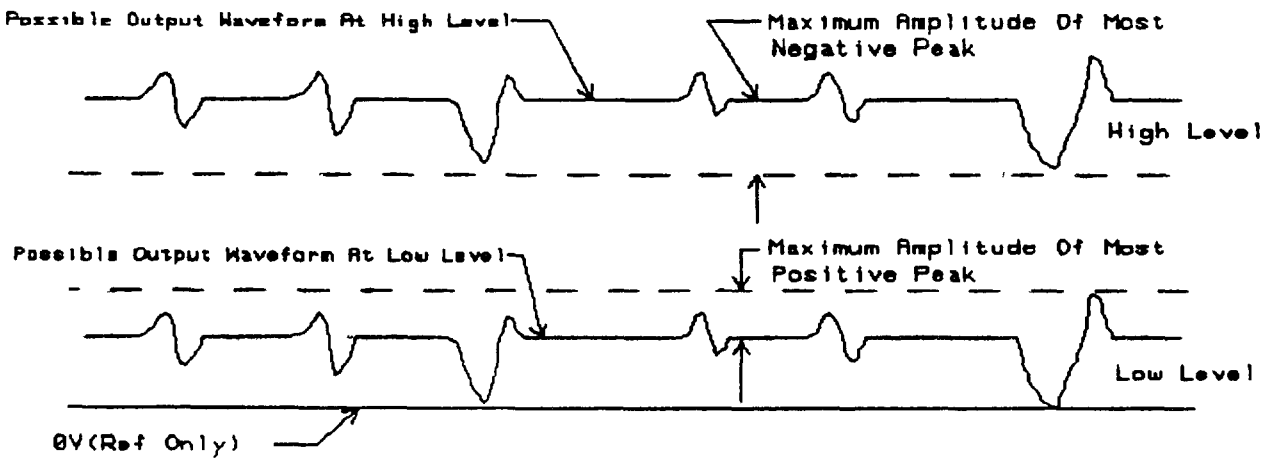


FIGURE 25. Output crosstalk (high level and low level) waveforms for tables XVII and XVIII.

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TABLE XVIII. Output crosstalk, high level, test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs
V1	Q1	21	0	400	mV	1-4	24,25
V2	Q2	70					
V3	Q3	20					
V4	Q4	23					
V5	Q5	72					
V6	Q6	22					
V7	Q7	71					
V8	1C	93					
V9	2C	42					
V10	3C	38					
V11	4C	90					
V12	5C	49					
V13	6C	98					
V14	7C	95					
V15	8C	46					
V16	ADDR/CONT DIR	68					
V17	DATA DIR	19					
V18	-ADDR/CONT EN	67					
V19	-DATA EN	18					
V20	-CTI BACK0	51	0	400	mV	1-4	24,25

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TABLE XVIII. Output crosstalk, high level, test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Figs
V21	-CTI BACK1	2	0	400	mV	1-4	24,25
V22	-CTI BACK2	52					
V23	-CTI BACK3	3					
V24	-CTI BACK4	53					
V25	-CTI BACK5	4					
V26	-CTI BR	54					
V27	-CTI BGACK	5					
V28	-I/O BACK0	26					
V29	-I/O BACK1	27					
V30	-I/O BACK2	77					
V31	-I/O BACK3	28					
V32	-I/O BACK4	78					
V33	-I/O BACK5	79					
V34	-REM BG A-B	30					
V35	-I/O BR	80					
V36	PU1	74					
V37	PU2	75	0	400	mV	1-4	24,25

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NOTES:

1. Output crosstalk, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 0.25 V	V1-V37
Power supply current limit	0.724 A	NA	V1-V37
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 V1-V37 V1-V37
Output crosstalk measurement	NA	± 20 mV	V1-V37

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) Use probe 'A' to verify the pattern voltage and pattern repetition rate of 1000 nS ± 200 nS, 50 percent duty cycle.

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NOTES: - Continued.

(d) For each output under test in table XVIII, connect probe 'B' and visually inspect the output for the most negative spike as indicated on figure 25.

(e) The maximum probe capacitance is 10 pF.

3. All outputs shall be loaded as specified on figure 14.

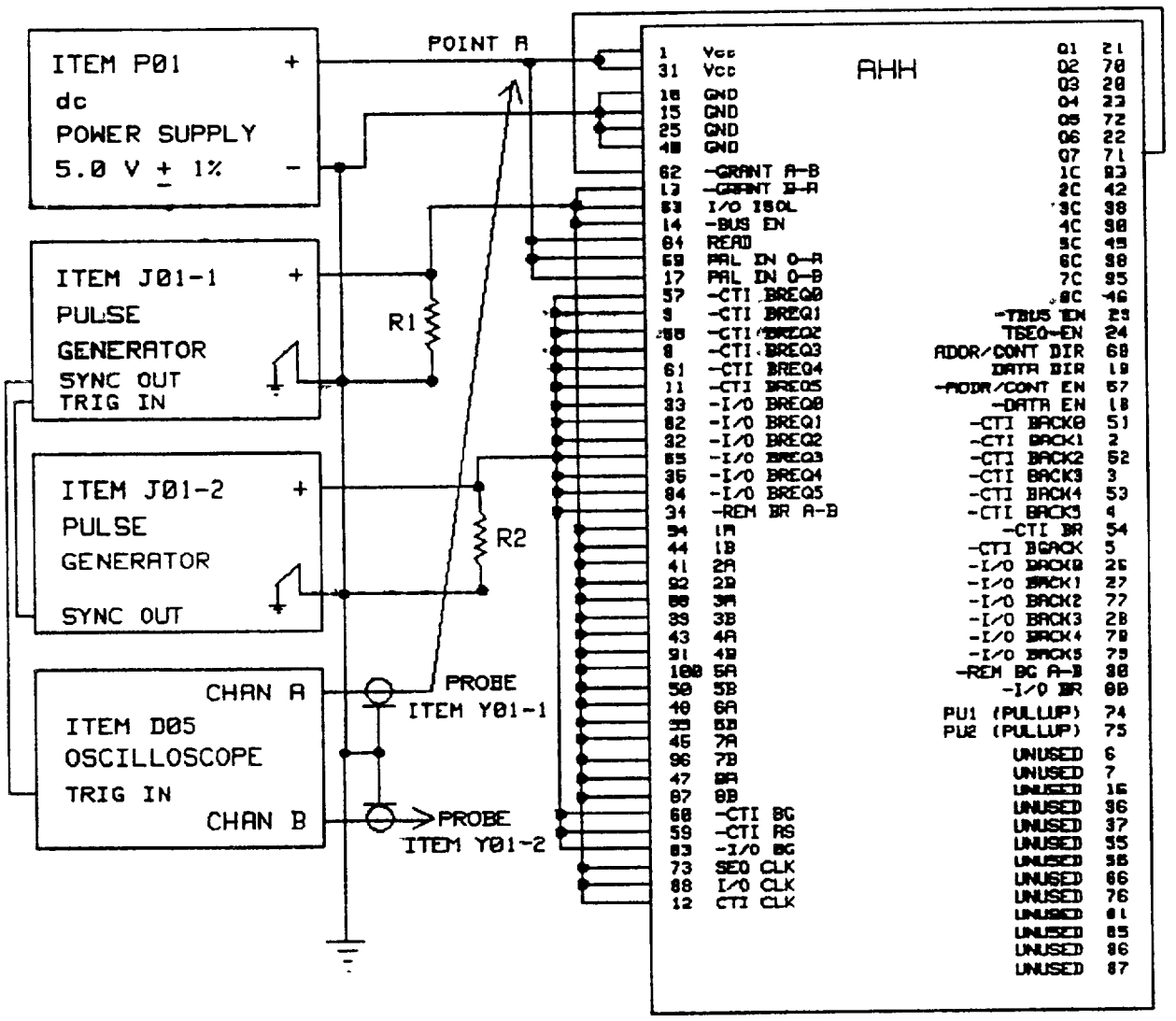
4. Abbreviations used are defined in 6.1.

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TABLE XIX. Power supply transient amplitude test.

Test cond	Parameter	25 ± 5°C Initial limits		Units	Test fig
		Min	Max		
W	PSTA	0	400	mV	26

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NOTE:

1. R1 = R2 = 51 Ohm ± 5 percent.

FIGURE 26. Power supply transient amplitude test circuit for table XIX.

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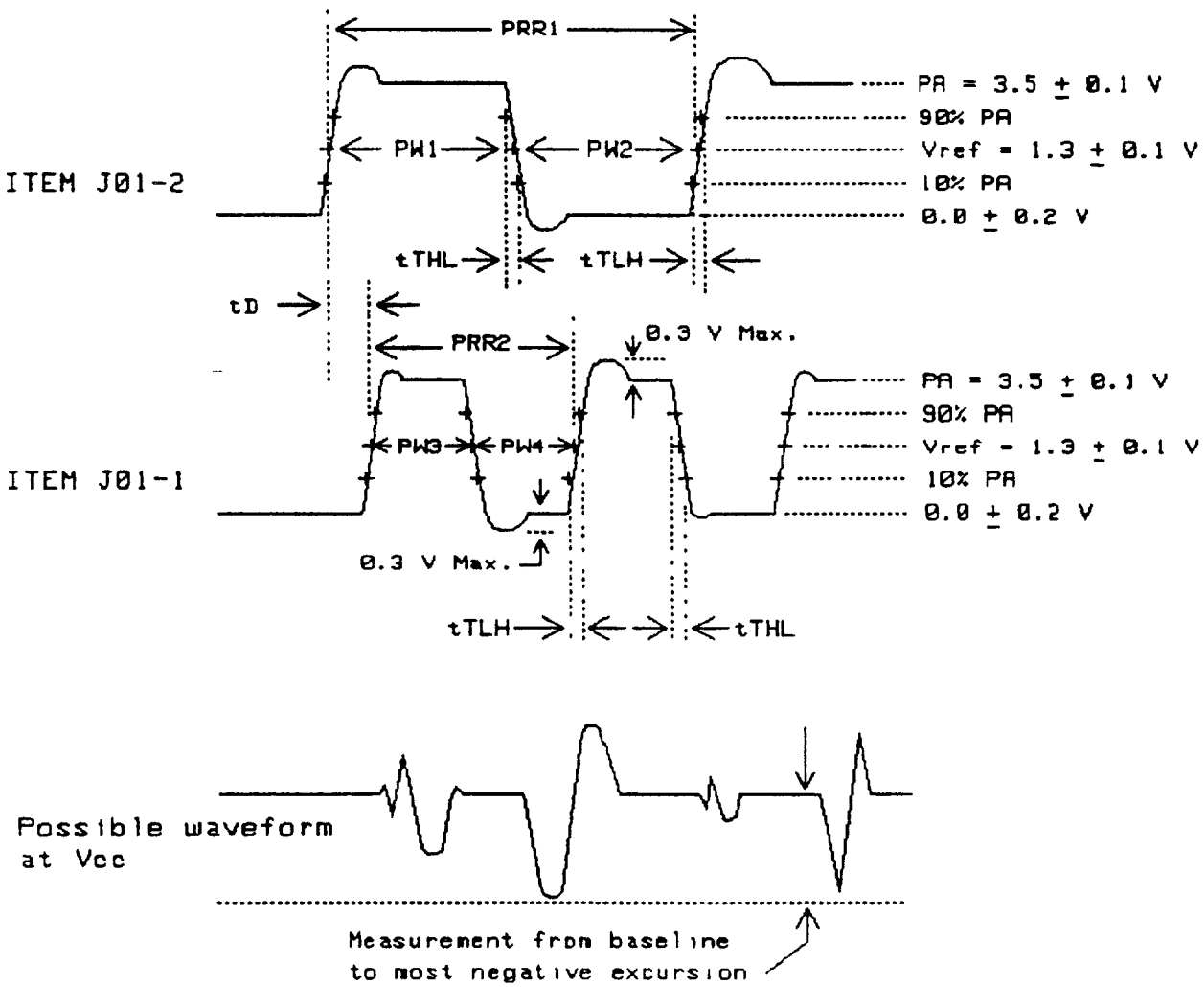


FIGURE 27. Power supply transient amplitude waveform for table XIX.

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TABLE XX. Power supply filter capacitance test.

Test cond	Parameter	B to pins	25 ± 5°C Initial limits Min	Max	Units	Test fig
X	PSFC	1,31	4.125	6.875	uF	28

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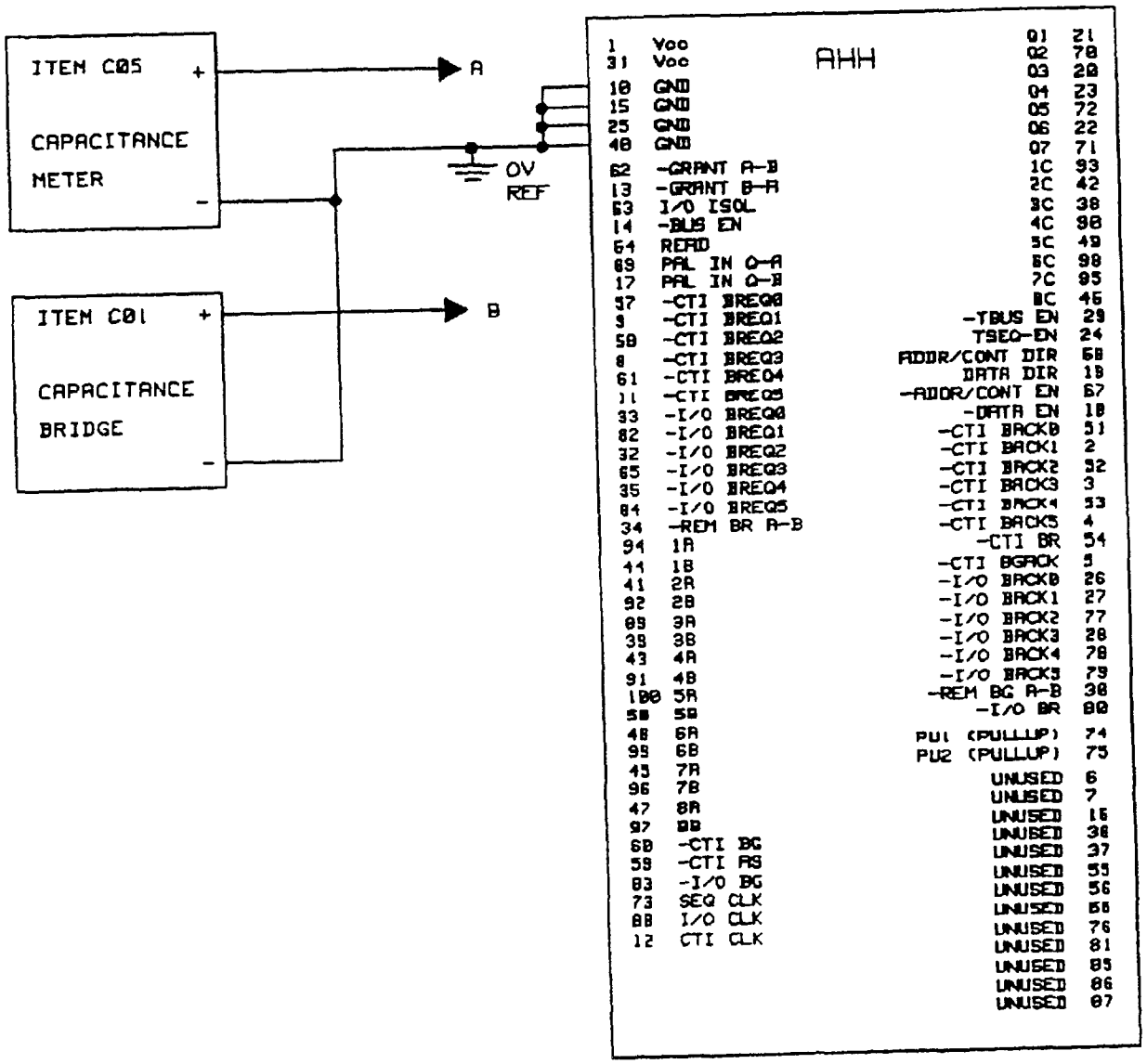


FIGURE 28. Power supply filter and input capacitance test circuit for table XX.

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TABLE XXI. Input capacitance test.

Test cond	Signal name	Connect A to pin	25 ± 5°C Initial limits Min	Max	Units	Fig
Y1	-GRANT A-B	62	2.0	55.0	pF	28
Y2	-GRANT B-A	13				
Y3	I/O ISOL	63				
Y4	-BUS EN	14	2.0	55.0	pF	28
Y5	READ	64	2.0	40.0	pF	28
Y6	PAL IN Q-A	69				
Y7	PAL IN Q-B	17	2.0	40.0	pF	28
Y8	-CTI BREQ0	57	2.0	55.0	pF	28
Y9	-CTI BREQ1	9				
Y10	-CTI BREQ2	58				
Y11	-CTI BREQ3	8				
Y12	-CTI BREQ4	61				
Y13	-CTI BREQ5	11				
Y14	-I/O BREQ0	33				
Y15	-I/O BREQ1	82				
Y16	-I/O BREQ2	32				
Y17	-I/O BREQ3	65				
Y18	-I/O BREQ4	35				
Y19	-I/O BREQ5	84				
Y20	-REM BR A-B	34	2.0	55.0	pF	28
Y21	1A	94	2.0	40.0	pF	28

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TABLE XXI. Input capacitance test - Continued.

Test cond	Signal name	Connect A to pin	25 ± 5°C Initial limits Min	Max	Units	Fig
Y22	1B	44	2.0	40.0	pF	28
Y23	2A	41				
Y24	2B	92				
Y25	3A	89				
Y26	3B	39				
Y27	4A	43				
Y28	4B	91				
Y29	5A	100				
Y30	5B	50				
Y31	6A	48				
Y32	6B	99				
Y33	7A	45				
Y34	7B	96				
Y35	8A	47				
Y36	8B	97				
Y37	-CTI BG	60				
Y38	-CTI AS	59				
Y39	-I/O BG	83	2.0	40.0	pF	28
Y40	SEQ CLK	73	2.0	30.0	pF	28
Y41	I/O CLK	88	2.0	70.0	pF	28
Y42	CTI CLK	12	2.0	70.0	pF	28

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TABLE XXII. Threshold test.

Test cond	Parameter	Notes	Fig
Z	Threshold	1-3	21

NOTES:

1. Threshold test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	Z
Power supply current limit	0.724 A	NA	Z
Pattern voltage high, (VH) low, (VL)	 2.0 V 0.7 V	 ± 20 mV ± 7 mV	 Z Z
Output voltage measurement high, (Voh) low, (Vol)	 NA NA	 ± 23.1 mV ± 5.5 mV	 Z Z

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test and verify the output patterns by monitoring the outputs and comparing the results to the output low and high levels specified in the output voltage tests. The module inputs shall not violate the minimum input timing parameters referenced on figure 21.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

3. Abbreviations used are defined in 6.1.

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APPENDIX

10. SCOPE

10.1 Scope. This appendix contains information regarding the module function. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

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APPENDIX

30. Module function. This module provides 6 levels of bus arbitration on the counter, timer and interrupt (CTI) bus and 7 levels of bus arbitration on the I/O bus. It also generates a DMA (FUM module-AM9516 based) to micropocessor (NAH module-68000 based) interface protocal.

30.1 CTI bus arbitration. The AHH module can arbitrate as many as six (6) DMA devices in the counter timer and interface (CTI) section. The device that is the highest priority will gain bus mastership when the device using the bus relinquishes it. The CTI section also provides AM9516 based (FUM module) to 68000 based (NAH module) interface protocol. A bus grant (-CTI BG) is accepted from the 68000 which enables the requesting DMA device to gain control of the bus. Bus request (-CTI BR) and bus grant acknowledge (-CTI BGACK) are supplied to the 68000. The AHH can arbitrate as many as six (6) DMA devices and one (1) remote bus request from CTI in the I/O section. The remote request (-REM BR A-B) has the highest priority. The highest priority pending request will always gain control of the bus when the bus is relinquished. Pull-up resistors are used on request inputs to insure that no request is inadvertantly made when an input is not used. (Attain a "1" when an input floats.)

30.1.1 Additional logic. Additional user logic can be implemented with the spare logic AND and OR gates provided on the module.

30.2 PAL timing. A shift register (delay register) is used in conjunction with the logic PAL to provide proper timing sequence for the interface of the CTI and I/O bus. Once the shift register has been enabled, each output (Q1-Q7) will consequently go high on each rising clock edge. The Q1-Q7 outputs must be backplane wired to the PAL IN Q- A/B inputs.

30.3 Arbitration PAL. The arbitration scheme in both the I/O and CTI sections are accomplished with the use of a Programmable Array Logic (PAL16R8).

30.4 Control PAL. A PAL (PAL12L6) is used on the module to implement certain logic functions. The enabling and direction control of bidirectional control buffers in the address/read partition buffer are controlled by the PAL12L6 outputs. The PAL also controls the enabling of the shift register and the arbiter PALs on the module.

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APPENDIX

30.5 Miscellaneous. The AHH module includes two pull-up resistors (10 KOhm) whose outputs (PU1,PU2) are pinned out for general purpose use. AND gate inputs (1 A/B to 4 A/B) which are not used should be left open. Tying LS gate inputs together significantly increases the gate's input noise susceptibility.

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APPENDIX

AHH PAL BOOLEANS

CTI BUS ARBITER LOGIC

-CTI BR(N+2) = $\overline{(-CTI\ BGACK(N+1))}$ + [(-CTI BREQ0(N)) x (-CTI BREQ1(N)) x (-CTI BREQ2(N)) x (-CTI BREQ3(N)) x (-CTI BREQ4(N)) x (-CTI BREQ5(N))]

-CTI BGACK(N+2) = [(-CTI BG(N)) + $\overline{(-CTI\ AS(N))}$] x [(-CTI BREQ0(N)) + (-CTI BACK0(N+1))] x [(-CTI BREQ1(N)) + [-CTI BACK1(N+1))] x [(-CTI BREQ2(N)) + (-CTI BACK2(N+1))] x [(-CTI BREQ3(N)) + (-CTI BACK3(N+1))] x [(-CTI BREQ4(N)) + (-CTI BACK4(N+1))] x [(-CTI BACK5(N)) + (-CTI BACK5(N+1))]

-CTI BACK0(N+2) = [(-CTI BREQ0(N)) + $\overline{(-CTI\ BREQ1(N))}$ + $\overline{(-CTI\ BREQ2(N))}$ + (-CTI BREQ3(N)) + (-CTI BREQ4(N)) + (-CTI BREQ5(N)) + (-CTI BACK5(N+1)) + (-CTI BACK4(N+1)) + (-CTI BACK3(N+1)) + (-CTI BACK2(N+1)) + (-CTI BACK1(N+1))] + (-CTI BGACK(n+1))] x [(-CTI BREQ0(N)) + (-CTI BACK0(N+1))]

-CTI BACK1(N+2) = [(-CTI BREQ1(N)) + $\overline{(-CTI\ BREQ2(N))}$ + $\overline{(-CTI\ BREQ3(N))}$ + (-CTI BREQ4(N)) + (-CTI BREQ5(N)) + (-CTI BACK5(N+1)) + (-CTI BACK4(N+1)) + (-CTI BACK3(N+1)) + (-CTI BACK2(N+1)) + (-CTI BACK0(N+1)) + (-CTI BGACK(N+1))] x [(-CTI BREQ1(N)) + (-CTI BACK1(N+1))]

-CTI BACK2(N+2) = [(-CTI BREQ2(N)) + $\overline{(-CTI\ BREQ3(N))}$ + $\overline{(-CTI\ BREQ4(N))}$ + (-CTI BREQ5(N)) + (-CTI BACK5(N+1)) + (-CTI BACK4(N+1)) + (-CTI BACK3(N+1)) + (-CTI BACK1(N+1)) + (-CTI BACK0(N+1)) + (-CTI BGACK(N+1))] x [(-CTI BREQ2(N)) + (-CTI BACK2(N+1))]

-CTI BACK3(N+2) = [(-CTI BREQ3(N)) + $\overline{(-CTI\ BREQ4(N))}$ + $\overline{(-CTI\ BREQ5(N))}$ + (-CTI BACK5(N+1)) + (-CTI BACK4(N+1)) + (-CTI BACK2(N+1)) + (-CTI BACK1(N+1)) + (-CTI BACK0(N+1)) + (-CTI BGACK(N+1))] x [(-CTI BREQ3(N)) + (-CTI BACK3(N+1))]

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APPENDIX

$$-\text{CTI BACK4}(N+2) = [(-\text{CTI BREQ4}(N)) + \overline{(-\text{CTI BREQ5}(N))} + \overline{(-\text{CTI BACK5}(N+1))} + \overline{(-\text{CTI BACK3}(N+1))} + \overline{(-\text{CTI BACK2}(N+1))} + \overline{(-\text{CTI BACK1}(N+1))} + \overline{(-\text{CTI BACK0}(N+1))} + (-\text{CTI BGACK}(N+1)) \times [(-\text{CTI BREQ4}(N)) + (-\text{CTI BACK4}(N+1))]$$

$$-\text{CTI BACK5}(N+2) = [(-\text{CTI BREQ5}(N)) + \overline{(-\text{CTI BACK4}(N+1))} + \overline{(-\text{CTI BACK3}(N+1))} + \overline{(-\text{CTI BACK2}(N+1))} + \overline{(-\text{CTI BACK1}(N+1))} + \overline{(-\text{CTI BACK0}(N+1))} + (-\text{CTI BGACK}(N+1))] \times [(-\text{CTI BREQ5}(N)) + (-\text{CTI BACK5}(N+1))]$$

NOTES:

1. XX(N) = Signal XX at state N
   XX(N+1) = Signal XX at state N+1
   XX(N+2) = Signal XX at state N+2

2. + = Logical OR operation, outside of parentheses.
   x = Logical AND operation.

Transition between states is determined by the rising edge of the input signal CTI CLK.

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APPENDIX

AHH PAL PARTITION

CTI - I/O PARTITION

BUFFER CONTROL LOGIC

TSEQ-EN = $(\overline{\text{-GRANT A-B}}) + (\overline{\text{-GRANT B-A}})$

-ADDR/CONT EN = [(-GRANT B-A) + I/O ISOL + $\overline{\text{PAL IN Q-A}}$ + (-BUS EN)]

[(-GRANT A-B) + I/O ISOL + PAL IN Q-A + (-BUS EN)]

-DATA EN = [(-GRANT B-A) + I/O ISOL + $\overline{\text{PAL IN Q-B}}$ + (-BUS EN)]

[(-GRANT A-B) + I/O ISOL + $\overline{\text{PAL IN Q-B}}$ + (-BUS EN)]

ADDR/CONT DIR = $\overline{\text{-GRANT B-A}}$

DATA DIR = [$\overline{\text{(-GRANT B-A)}}$ + READ] [(-GRANT B-A) + $\overline{\text{READ}}$]

-TBUS EN = "LOGIC 0"

NOTE:

1. + = Logical OR operation.

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APPENDIX

AHH PAL BOOLEANS

I/O BUS ARBITER LOGIC

$$-\text{I/O BR}(N+2) = (-\text{I/O BREQ0}(N)) \times (-\text{I/O BREQ1}(N)) \times (-\text{I/O BREQ2}(N)) \times (-\text{I/O BREQ3}(N)) \times (-\text{I/O BREQ4}(N)) \times (\text{I/O BREQ5}(N)) \times (-\text{REM BR A-B}(N))$$

$$-\text{I/O BACK0}(N+2) = \overline{[(-\text{I/O BREQ0}(N)) + \overline{(-\text{I/O BREQ1}(N)) + \overline{(-\text{I/O BREQ2}(N)) + \overline{(-\text{I/O BREQ3}(N)) + \overline{(-\text{I/O BREQ4}(N)) + \overline{(-\text{I/O BREQ5}(N))} + \overline{(-\text{REM BR A-B}(N) + \overline{(-\text{I/O BACK5}(N+1)) + \overline{(-\text{I/O BACK4}(N+1)) + \overline{(-\text{I/O BACK3}(N+1)) + \overline{(-\text{I/O BACK2}(N+1)) + \overline{(-\text{I/O BACK1}(N+1)) + \overline{(-\text{REM BG A-B}(N+1)) + (-\text{I/O BG}(N))}}}

APPENDIX

$$-\text{I/O BACK5}(N+2) = [(-\text{I/O BREQ5}(N)) + (\overline{-\text{REM BR A-B}(N)}) + (\overline{-\text{I/O BACK4}(N+1)}) + (\overline{-\text{I/O BACK3}(N+1)}) + (\overline{-\text{I/O BACK2}(N+1)}) + (\overline{-\text{I/O BACK1}(N+1)}) + (\overline{-\text{I/O BACK0}(N+1)}) + (-\text{REM BG A-B}(N+1)) + (-\text{I/O BG}(N))] [(-\text{I/O BREQ5}(N)) + (-\text{I/O BACK5}(N+1))]$$

$$-\text{REM BG A-B}(N+2) = [(-\text{REM BR A-B}(N)) + (\overline{-\text{I/O BACK5}(N+1)}) + (\overline{-\text{I/O BACK4}(N+1)}) + (-\text{I/O BACK3}(N+1)) + (-\text{I/O BACK2}(N+1)) + (-\text{I/O BACK1}(N+1)) + (-\text{I/O BACK0}(N+1)) + (-\text{I/O BG}(N))] [(-\text{REM BR A-B}(N)) + (-\text{REM BG A-B}(N+1))]$$

NOTES:

1. XX(N) = Signal XX at state N
   XX(N+1) = Signal XX at state N+1
   XX(N+2) = Signal XX at state N+2

2. + = Logical OR operation, outside of parentheses.
   x = Logical AND operation.

Transition between states is determined by the rising edge of the input signal I/O CLK.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

The preparing activity must complete blocks 1, 2, 3, and 8 In block 1, both the document number and revision letter should be given

2 The submitter of this form must complete blocks 4, 5, 6, and 7

3 The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1 DOCUMENT NUMBER MIL-M-28787/356	2 DOCUMENT DATE (YYMMDD)

3 DOCUMENT TITLE
Modules, Digital, Standard Electronic, Support, MPU Communications Module, Key Code AHH

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible Attach extra sheets as needed.)

5 REASON FOR RECOMMENDATION

6 SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a NAME Technical Point of Contact (TPOC): Mr. Robert F. Dilger (Code 602A) ASE ADDRESS ALL CORRESPONDENCE AS FOLLOWS:	b TELEPHONE (Include Area Code) (1) Commercial: TPOC: 812-854-1528 (2) AUTOVON: 88-482-1528
c ADDRESS (Include Zip Code) Commanding Officer Naval Weapons Support Center ATTN: Robert F. Dilger, Bldg. 2940W, Code 602A Crane, Indiana 47522-5060	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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